

## NCERT Exemplar Solutions for Class 12

### Mathematics

#### Chapter 3 - Matrices

#### Short Answer:

#### Sample questions:

1. Construct a matrix  $A = [a_{ij}]_{2 \times 2}$  whose elements are given by  $a_{ij} = e^{2ix} \sin jx$ .

**Ans: Given:** Elements  $a_{ij} = e^{2ix} \sin jx$ .

Form the elements by substituting values of  $i$  and  $j$ . Then, construct a matrix.

For the values of  $i$  and  $j$ .

$$i = 1, j = 1 \text{ gives } a_{11} = e^{2x} \sin x$$

$$i = 1, j = 2 \text{ gives } a_{12} = e^{2x} \sin 2x$$

$$i = 2, j = 1 \text{ gives } a_{21} = e^{4x} \sin x$$

$$i = 2, j = 2 \text{ gives } a_{22} = e^{4x} \sin 2x$$

$$\therefore A = \begin{bmatrix} e^{2x} \sin x & e^{2x} \sin 2x \\ e^{4x} \sin x & e^{4x} \sin 2x \end{bmatrix}$$

2. If  $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 3 & 1 \end{bmatrix}$ ,  $C = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ ,  $D = \begin{bmatrix} 4 & 6 & 8 \\ 5 & 7 & 9 \end{bmatrix}$  then which of the sums  $A + B$ ,  $B + C$ ,  $C + D$  and  $B + D$  is defined?

**Ans: Given:** Matrices  $A$ ,  $B$  and  $C$ .

The sum of two matrices is defined only when they have same order.

Since, matrices having same order can be added.

$$\text{Order of } A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}_{2 \times 2} \text{ and } B = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 3 & 1 \end{bmatrix}_{2 \times 3},$$

$$\text{Order of } B = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 3 & 1 \end{bmatrix}_{2 \times 3} \text{ and } C = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2 \times 1},$$

$$\text{Order of } C = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2 \times 1} \text{ and } D = \begin{bmatrix} 4 & 6 & 8 \\ 5 & 7 & 9 \end{bmatrix}_{2 \times 3},$$

$$\text{Order of } B = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 3 & 1 \end{bmatrix}_{2 \times 3} \text{ and } D = \begin{bmatrix} 4 & 6 & 8 \\ 5 & 7 & 9 \end{bmatrix}_{2 \times 3}.$$

Therefore,  $B + D$  with order  $2 \times 3$  is defined.

**3. Show that a matrix which is both symmetric and skew symmetric is a zero matrix.**

**Ans: Given:** Symmetric and Skew symmetric matrix.

For symmetric matrix  $A' = A$  and for skew symmetric matrix  $A' = -A$ .

Let  $A$  be a matrix with elements  $a_{ij}$ .

As,  $A$  is both symmetric and skew symmetric,

$$\Rightarrow a_{ij} = a_{ji} \quad (i)$$

$$\Rightarrow a_{ij} = -a_{ji}$$

$$\Rightarrow a_{ij} + a_{ji} = 0 \quad (ii)$$

from (i) and (ii),

$$2a_{ij} = 0$$

$$\therefore a_{ij} = 0$$

Therefore, A is zero matrix.

4. If  $\begin{bmatrix} 2x & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix} \begin{bmatrix} x \\ 8 \end{bmatrix} = 0$ , find the value of x.

**Ans: Given:**  $\begin{bmatrix} 2x & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix} \begin{bmatrix} x \\ 8 \end{bmatrix} = 0$ .

Use matrix multiplication. Then compare the elements with the zero matrix.

Here,

$$\begin{bmatrix} 2x & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix} \begin{bmatrix} x \\ 8 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 2x - 9 & 4x \end{bmatrix} \begin{bmatrix} x \\ 8 \end{bmatrix} = [0]$$

$$\Rightarrow [2x^2 - 9x + 32x] = [0]$$

$$\Rightarrow 2x^2 - 23x = 0$$

$$\Rightarrow x(2x + 23) = 0$$

$$\Rightarrow x = 0, x = \frac{-23}{2}$$

5. If A is  $3 \times 3$  invertible matrix, then show that for any scalar k(non zero), kA is invertible and  $(kA)^{-1} = \frac{1}{k} A^{-1}$ .

**Ans: Given:** A is  $3 \times 3$  invertible matrix.

For invertible matrix,  $AA^{-1} = I$ .

Here,

$$(kA)\left(\frac{1}{k}A^{-1}\right) = \left(k \cdot \frac{1}{k}\right)(AA^{-1})$$

$$\Rightarrow 1 \cdot I = I$$

$$\therefore (kA) \text{ is inverse of } \left(\frac{1}{k}A^{-1}\right).$$

**Long Answer:**

**6. Express the matrix A as the sum of a symmetric and a skew symmetric matrix, where**

$$A = \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}.$$

**Ans: Given:** Matrix A.

The matrix A is expressed as  $\frac{A + A'}{2} + \frac{A - A'}{2}$ . Here

$\frac{A + A'}{2}$  is symmetric and  $\frac{A - A'}{2}$  is skew-symmetric matrix.

Here,

$$A = \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}, \text{ then } A' = \begin{bmatrix} 2 & 7 & 1 \\ 4 & 3 & -2 \\ -6 & 5 & 4 \end{bmatrix}$$

$$\text{now, } \frac{A + A'}{2} = \frac{1}{2} \begin{bmatrix} 4 & 11 & -5 \\ 11 & 6 & 3 \\ -5 & 3 & 8 \end{bmatrix}$$

$$\Rightarrow \frac{A + A'}{2} = \begin{bmatrix} 2 & \frac{11}{2} & \frac{-5}{2} \\ \frac{11}{2} & 3 & \frac{3}{2} \\ \frac{-5}{2} & \frac{3}{2} & 4 \end{bmatrix}$$

$$\text{now, } \frac{A - A'}{2} = \frac{1}{2} \begin{bmatrix} 0 & -3 & -7 \\ 3 & 0 & 7 \\ 7 & -7 & 0 \end{bmatrix}$$

$$\Rightarrow \frac{A - A'}{2} = \begin{bmatrix} 0 & \frac{-3}{2} & \frac{-7}{2} \\ \frac{3}{2} & 0 & \frac{7}{2} \\ \frac{7}{2} & \frac{-7}{2} & 0 \end{bmatrix}$$

$$\therefore \frac{A + A'}{2} + \frac{A - A'}{2} = \begin{bmatrix} 2 & \frac{11}{2} & \frac{-5}{2} \\ \frac{11}{2} & 3 & \frac{3}{2} \\ \frac{-5}{2} & \frac{3}{2} & 4 \end{bmatrix} + \begin{bmatrix} 0 & \frac{-3}{2} & \frac{-7}{2} \\ \frac{3}{2} & 0 & \frac{7}{2} \\ \frac{7}{2} & \frac{-7}{2} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}$$

$$= A.$$

7. If  $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$ , then show that A satisfies the equation

$$A^3 - 4A^2 - 3A + 11I = 0.$$

**Ans: Given:** Matrix A.

First, find  $A^3$  and  $A^2$ . Then, substitute in the equation.

Here,

$$A^2 = A \times A$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1+6+2 & 3+0+4 & 2-3+6 \\ 2+0-1 & 6+0-2 & 4+0-3 \\ 1+4+3 & 3+0+6 & 2-2+9 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix}$$

$$\text{Now, } A^3 = A^2 \times A$$

$$\Rightarrow A^3 = \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} \times \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\Rightarrow A^3 = \begin{bmatrix} 9+14+5 & 27+0+10 & 18-7+15 \\ 1+8+1 & 3+0+2 & 2-4+3 \\ 8+18+9 & 24+0+18 & 16-9+27 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix}$$

Substitute the values in the equation,  $A^3 - 4A^2 - 3A + 11I = 0$ .

$$\begin{aligned}
 &= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} - 4 \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} - 3 \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} + 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} 28 - 36 - 3 + 11 & 37 - 28 - 9 + 0 & 26 - 20 - 6 + 0 \\ 10 - 4 - 6 + 0 & 5 - 16 + 0 + 11 & 1 - 4 + 3 + 0 \\ 35 - 32 - 3 + 0 & 42 - 36 - 6 + 0 & 34 - 36 - 9 + 11 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
 &= 0
 \end{aligned}$$

**8. Let  $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ . Then show that  $A^2 - 4A + 7I = 0$ . Using this result calculate  $A^5$  also.**

**Ans: Given:** Matrix  $A$ .

First, find  $A^2$  and use  $A^5 = A^3 A^2$ .

Here,

$$\begin{aligned}
 A^2 &= A \times A \\
 \Rightarrow A^2 &= \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \\
 \Rightarrow A^2 &= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix}
 \end{aligned}$$

Substitute in the equation,  $A^2 - 4A + 7I = 0$ .

$$= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - 4 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 - 8 + 7 & 12 - 12 + 0 \\ -4 + 4 + 0 & 1 - 8 + 7 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$= 0$$

$$\therefore A^2 - 4A + 7I = 0$$

$$\Rightarrow A^2 = 4A - 7I$$

Now,

$$A^3 = A.A^2$$

$$\Rightarrow A^3 = A.(4A - 7I)$$

$$\Rightarrow A^3 = 4(4A - 7I) - 7A$$

$$\Rightarrow A^3 = 16A - 28I - 7A$$

$$\Rightarrow A^3 = 9A - 28I$$

$$\text{Now, } A^5 = A^3 A^2$$

$$\Rightarrow A^5 = (9A - 28I)(4A - 7I)$$

$$\Rightarrow A^5 = 36A^2 - 63A - 112A + 196I$$

$$\Rightarrow A^5 = 36(4A - 7I) - 175A + 196I$$

$$\Rightarrow A^5 = -31A - 56I$$

$$\Rightarrow A^5 = -31 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 56 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^5 = - \begin{bmatrix} 62 & 93 \\ -31 & 62 \end{bmatrix} - \begin{bmatrix} 56 & 0 \\ 0 & 56 \end{bmatrix}$$

$$\Rightarrow A^5 = \begin{bmatrix} -62 - 56 & -93 - 0 \\ 31 & -62 - 56 \end{bmatrix}$$

$$\Rightarrow A^5 = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix}$$

### Objective Type Questions:

Choose the correct answer from the given four options in Examples 9 to 12.

9. If  $A$  and  $B$  are matrices of same order, then  $(A + B)(A - B)$  is equal to

(A)  $A^2 - B^2$

(B)  $A^2 - BA - AB - B^2$

(C)  $A^2 - B^2 + BA - AB$

(D)  $A^2 - BA + B^2 + AB$

**Ans: Given:**  $A$  and  $B$  are matrices.

Use the matrices algebra.

Here,

$$(A + B)(A - B) = A(A - B) + B(A - B)$$

$$\Rightarrow (A + B)(A - B) = A^2 - AB + BA - B^2.$$

**Correct Answer:** C

10. If  $A = \begin{bmatrix} 2 & -1 & 3 \\ -4 & 5 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 3 \\ 4 & -2 \\ 1 & 5 \end{bmatrix}$ , then

- (A) Only  $AB$  is defined
- (B) Only  $BA$  is defined
- (C)  $AB$  and  $BA$  both are defined
- (D)  $AB$  and  $BA$  both are not defined

**Ans: Given:** Matrices  $A$  and  $B$ .

Product of two matrices is defined when column of one matrix is equal to the row of another matrix.

The order of matrix  $A$  is  $2 \times 3$  and order of  $B$  is  $3 \times 2$ .

Therefore, both  $AB$  and  $BA$  is defined.

**Correct Answer: C**

11. The matrix  $A = \begin{bmatrix} 0 & 0 & 5 \\ 0 & 5 & 0 \\ 5 & 0 & 0 \end{bmatrix}$  is a

- (A) Scalar matrix
- (B) Diagonal matrix
- (C) Unit matrix
- (D) Square matrix

**Ans: Given:** Matrix  $A$ .

Matrix can be square, diagonal, unit depending on the order and elements of matrix.

Since, the order of matrix  $A$  is  $3 \times 3$ . The number of rows and columns are same.

Therefore, matrix  $A$  is square matrix.

**Correct Answer:** D

**12. If  $A$  and  $B$  are symmetric matrices of the same order, then  $(AB' - BA')$  is a**

**(A) Skew symmetric matrix**

**(B) Null matrix**

**(C) Symmetric matrix**

**(D) None of these**

**Ans: Given:**  $A$  and  $B$  are symmetric matrices.

For symmetric matrix  $A' = A$  and for skew symmetric matrix  $A' = -A$ .

Take transpose,

$$(AB' - BA')' = (AB')' - (BA')'$$

$$(AB' - BA')' = (BA' - AB')$$

$$(AB' - BA')' = -(AB' - BA').$$

**Correct Answer:** A

**Fill in the blanks in each of the Examples 13 to 15.**

**13. If A and B are two skew symmetric matrices of same order, then AB is symmetric matrix if \_\_\_\_\_ .**

**Ans: Given:** A and B are two skew symmetric matrices.

For symmetric matrix  $A' = A$  and for skew symmetric matrix  $A' = -A$ .

Take transpose,

$$(AB)' = B'A'$$

$$\Rightarrow (AB)' = (-B)(-A)$$

$$(AB)' = BA$$

if,  $AB = BA$ , then

$$(AB)' = AB.$$

For  $AB = BA$ , AB is symmetric matrix.

**14. If A and B are matrices of same order, then  $(3A - 2B)'$  is equal to \_\_\_\_ .**

**Ans: Given:** A and B are matrices.

Take transpose of  $(3A - 2B)'$ .

Taking transpose,

$$(3A - 2B)' = 3A' - 2B'.$$

**15. Addition of matrices is defined if order of the matrices is \_\_\_\_.**

**Ans:** Same.

The order of must be same for addition of two matrices.

State whether the statements in each of the Examples 16 to 19 is true or false:

**16. If two matrices A and B are of same order, then  $2A + B = B + 2A$ .**

**Ans: Given:** A and B are matrices.

Matrix addition is defined only for the matrices with same order.

Since, A and B are of same order. The matrix addition is defined.

Therefore,  $2A + B = B + 2A$  is true.

**17. Matrix subtraction is associative.**

**Ans: Given:** Matrix.

Consider three matrices to verify the property of associativity.

Let A, B and C are matrices,

$$(A - B) - C = A - B - C \quad (i)$$

$$A - (B - C) = A - B + C \quad (ii)$$

$$(i) \neq (ii)$$

$\therefore$  False.

**18. For the non-singular matrix A,  $(A')^{-1} = (A^{-1})'$ .**

**Ans:** For a non-singular matrix  $|A| \neq 0$ . The identity  $A, (A')^{-1} = (A^{-1})'$  is always true.

**19.  $AB = AC$  &  $B = C$  for any three matrices of same order.**

**Ans: Given:** A, B and C are matrices.

It is not given that, matrix  $A$  is invertible.

It is not mentioned that matrix  $A$  is invertible.

$AB = AC \Rightarrow B = C$  is false for non-invertible matrix.

## EXERCISE:

### Short Answer Type:

**1. If a matrix has 28 elements, what are the possible orders it can have? What if it has 13 elements?**

**Ans: Given:** Number of elements in matrix.

The order of a matrix is represented by  $m \times n$ . The number of elements of order  $m \times n$  will be  $mn$ .

For,  $mn = 28$ ,

Possible ordered pairs,

$$(m \times n) = \{(1, 28), (2, 14), (4, 7), (7, 4), (14, 2), (28, 1)\}$$

for 28 elements possible orders  $1 \times 28, 2 \times 14, 4 \times 7, 7 \times 4, 14 \times 2, 28 \times 1$ .

for 13 elements possible orders  $1 \times 13$  and  $13 \times 1$ .

**2. In the matrix  $A = \begin{bmatrix} a & 1 & x \\ 2 & \sqrt{3} & x^2 - y \\ 0 & 5 & \frac{-2}{5} \end{bmatrix}$ , write**

**(i) The order of the matrix  $A$ .**

**Ans:** First, find the order  $m \times n$  and then number of elements using  $mn$ . General representation of element is  $a_{ij}$ .

Order of matrix  $A = 3 \times 3$ .

**(ii) The number of elements.**

**Ans:** First, find the order  $m \times n$  and then number of elements using  $mn$ . General representation of element is  $a_{ij}$ .

Number of elements  $3 \times 3 = 9$ .

**(iii) Elements  $a_{23}$ ,  $a_{31}$  and  $a_{12}$ .**

**Ans:** First, find the order  $m \times n$  and then number of elements using  $mn$ . General representation of element is  $a_{ij}$ .

$$a_{23} = x^2 - y, a_{31} = 0 \text{ and } a_{12} = 1.$$

**3. Construct  $a_{2 \times 2}$  matrix, where**

**(i)  $a_{ij} = \frac{(i - 2j)^2}{2}$**

**Ans: Given:** Matrix  $a_{2 \times 2}$ .

Substitute the values of  $i$  and  $j$  for  $2 \times 2$  matrix.

Here,

$$a_{ij} = \frac{(i - 2j)^2}{2},$$

for,

$$a_{11} = \frac{(1 - 2)^2}{2} = \frac{1}{2}$$

$$a_{12} = \frac{(1 - 2 \times 2)^2}{2} = \frac{9}{2}$$

$$a_{21} = \frac{(2 - 2 \times 1)^2}{2} = 0$$

$$a_{22} = \frac{(2 - 2 \times 2)^2}{2} = 2$$

$$\therefore A = \begin{bmatrix} \frac{1}{2} & \frac{9}{2} \\ 0 & 2 \end{bmatrix}$$

**(ii)  $a_{ij} = |-2i + 3j|$**

**Ans: Given:** Matrix  $a_{2 \times 2}$ .

Substitute the values of i and j for  $2 \times 2$  matrix.

Here,

$$a_{ij} = |-2i + 3j|,$$

for

$$a_{11} = |-2 \times 1 + 3 \times 1| = 1$$

$$a_{12} = |-2 \times 1 + 3 \times 2| = 4$$

$$a_{21} = |-2 \times 2 + 3 \times 1| = 1$$

$$a_{22} = |-2 \times 2 + 3 \times 2| = 2$$

$$\therefore A = \begin{bmatrix} 1 & 4 \\ 1 & 2 \end{bmatrix}$$

**4. Construct a  $3 \times 2$  matrix whose elements are given by  $a_{ij} = e^{ix} \sin jx$ .**

**Ans: Given:** Elements  $a_{ij} = e^{ix} \sin jx$ .

Form the elements by substituting values of i and j. Then, construct a matrix.

For the values of  $i$  and  $j$ .

$$i = 1, j = 1 \text{ gives } a_{11} = e^x \sin x$$

$$i = 1, j = 2 \text{ gives } a_{12} = e^x \sin 2x$$

$$i = 2, j = 1 \text{ gives } a_{21} = e^{2x} \sin x$$

$$i = 2, j = 2 \text{ gives } a_{22} = e^{2x} \sin 2x$$

$$i = 3, j = 1 \text{ gives } a_{31} = e^{3x} \sin x$$

$$i = 3, j = 2 \text{ gives } a_{32} = e^{3x} \sin 2x$$

$$\therefore A = \begin{bmatrix} e^x \sin x & e^x \sin 2x \\ e^{2x} \sin x & e^{2x} \sin 2x \\ e^{3x} \sin x & e^{3x} \sin 2x \end{bmatrix}$$

**5. Find the values of  $a$  and  $b$ , if  $A = B$ , where**

$$A = \begin{bmatrix} a + 4 & 3b \\ 8 & -6 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2a + 2 & b^2 + 2 \\ 8 & b^2 - 5b \end{bmatrix}$$

**Ans: Given:**  $A = B$ .

If two matrices are equal, then each element of  $A$  is equal to corresponding element of  $B$ .

Since,  $A = B$ ,

$$\begin{bmatrix} a + 4 & 3b \\ 8 & -6 \end{bmatrix} = \begin{bmatrix} 2a + 2 & b^2 + 2 \\ 8 & b^2 - 5b \end{bmatrix}$$

$$a_{11} = b_{11}$$

$$\Rightarrow a + 4 = 2a + 2$$

$$\Rightarrow a = 2$$

now,  $a_{12} = b_{12}$

$$\Rightarrow 3b = b^2 + 2$$

$$\Rightarrow b^2 = 3b - 2 \quad \text{-(i)}$$

and,  $a_{22} = b_{22}$

$$\Rightarrow -6 = b^2 - 5b \quad \text{-(ii)}$$

from (i) and (ii), we get

$$-6 = 3b - 2 - 5b$$

$$\Rightarrow b = 2$$

$$\therefore a = 2 \text{ and } b = 2.$$

**6. If possible, find the sum of matrices A and B, where**

$$A = \begin{bmatrix} \sqrt{3} & 1 \\ 2 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} x & y & z \\ a & b & c \end{bmatrix}.$$

**Ans: Given:** Matrices A and B.

Two matrices are added, if they have same order.

Since, matrices A and B are of different order. The order of matrices A and B are  $2 \times 2$  and  $2 \times 3$  respectively.

Therefore, the sum of matrices A and B is not possible.

**7. If  $X = \begin{bmatrix} 3 & 1 & -1 \\ 5 & -2 & -3 \end{bmatrix}$  and  $Y = \begin{bmatrix} 2 & 1 & -1 \\ 7 & 2 & 4 \end{bmatrix}$ , then find**

**(i)  $X + Y$ .**

**Ans: Given:** X and Y are matrices.

Here,

$$X + Y = \begin{bmatrix} 3+2 & 1+1 & -1-1 \\ 5+7 & -2+2 & -3+4 \end{bmatrix}$$

$$\therefore X + Y = \begin{bmatrix} 5 & 2 & -2 \\ 12 & 0 & 1 \end{bmatrix}$$

**(ii)  $2X - 3Y$ .**

**Ans: Given:** X and Y are matrices.

Here,

$$2X - 3Y = 2 \begin{bmatrix} 3 & 1 & -1 \\ 5 & -2 & -3 \end{bmatrix} - 3 \begin{bmatrix} 2 & 1 & -1 \\ 7 & 2 & 4 \end{bmatrix}$$

$$2X - 3Y = \begin{bmatrix} 6-6 & 2-3 & -2+3 \\ 10-21 & -4-6 & -6-12 \end{bmatrix}$$

$$\therefore 2X - 3Y = \begin{bmatrix} 0 & -1 & 1 \\ -11 & -10 & -18 \end{bmatrix}$$

**(iii) A matrix Z such that  $X + Y + Z$  is a zero matrix.**

**Ans: Given:** X and Y are matrices.

Here,

$$X + Y = \begin{bmatrix} 3+2 & 1+1 & -1-1 \\ 5+7 & -2+2 & -3+4 \end{bmatrix}$$

$$X + Y = \begin{bmatrix} 5 & 2 & -2 \\ 12 & 0 & 1 \end{bmatrix}$$

Since,  $X + Y + Z = 0$

$$\Rightarrow Z = -(X + Y)$$

$$\therefore Z = \begin{bmatrix} -5 & -2 & 2 \\ -12 & 0 & -1 \end{bmatrix}$$

**8. Find non-zero values of  $x$  satisfying the matrix equation**

$$x \begin{bmatrix} 2x & 2 \\ 3 & x \end{bmatrix} + 2 \begin{bmatrix} 8 & 5x \\ 4 & 4x \end{bmatrix} = 2 \begin{bmatrix} (x^2 + 8) & 24 \\ 10 & 6x \end{bmatrix}.$$

**Ans: Given:** Matrix equation

Solve the matrix operations. Then, compare the elements of matrices.

Here,

$$\begin{aligned} x \begin{bmatrix} 2x & 2 \\ 3 & x \end{bmatrix} + 2 \begin{bmatrix} 8 & 5x \\ 4 & 4x \end{bmatrix} &= 2 \begin{bmatrix} (x^2 + 8) & 24 \\ 10 & 6x \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 2x^2 & 2x \\ 3x & x^2 \end{bmatrix} + \begin{bmatrix} 16 & 10x \\ 8 & 8x \end{bmatrix} &= \begin{bmatrix} 2x^2 + 16 & 48 \\ 20 & 12x \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 2x^2 + 16 & 2x + 10x \\ 3x + 8 & x^2 + 8x \end{bmatrix} &= \begin{bmatrix} 2x^2 + 16 & 48 \\ 20 & 12x \end{bmatrix} \\ \Rightarrow 2x + 10x &= 48 \\ \Rightarrow x &= \frac{48}{12} \\ \therefore x &= 4 \end{aligned}$$

**9. If  $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ , then show that  $(A + B)(A - B)^1 A^2 - B^2$ .**

**Ans: Given:** Matrices  $A$  and  $B$ .

Substitute matrices in equation. Then use matrix algebra.

Here,

$$A + B = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix}$$

$$\therefore (A + B)_{2 \times 2} \cdot (A - B)_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 5 \end{bmatrix} \quad \dots(i)$$

$$\text{Now, } A^2 = A \cdot A$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 0+1 & 0+1 \\ 0+1 & 1+1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\text{Now, } B^2 = B \cdot B$$

$$\Rightarrow B^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow B^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\therefore A^2 - B^2 = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \quad \dots(ii)$$

From equation (i) and (ii), we get

$$(A + B)(A - B) \neq A^2 - B^2$$

**10. Find the value of x, if**  $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = 0.$

**Ans: Given:** Matrix equation.

Product of matrix is possible if number of rows of matrix is equal to number of columns of another matrix.

$$\begin{bmatrix} 1 & x & 1 \end{bmatrix}_{1 \times 3} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix}_{3 \times 1} = 0.$$

$$\Rightarrow \begin{bmatrix} 1 + 2x + 15 & 3 + 5x + 3 & 2 + x + 2 \end{bmatrix}_{1 \times 3} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix}_{3 \times 1} = 0$$

$$\Rightarrow \begin{bmatrix} 16 + 2x & 5x + 6 & 2 + x + 2 \end{bmatrix}_{1 \times 3} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix}_{3 \times 1} = 0$$

$$\Rightarrow [16 + 2x + 2.(5x + 6) + (x + 4).x]_{1 \times 1} = 0$$

$$\Rightarrow [x^2 + 16x + 28]_{1 \times 1} = 0$$

$$\Rightarrow x^2 + 16x + 28 = 0$$

$$\Rightarrow (x + 2)(x + 14) = 0$$

$$\Rightarrow x = -2, -14$$

**11. Show that**  $A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$  **satisfies the equation**  $A^2 - 3A - 7I = 0$  **and hence find the value of**  $A^{-1}$ .

**Ans: Given:** Matrix A.

First, find  $A^2$ . Then, substitute the values in the equation.

Here,

$$A^2 = A.A$$

$$\Rightarrow A^2 = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix}$$

$$\text{Now, } A^2 - 3A - 7I,$$

$$= \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix} - 3 \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} - 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 22 - 15 - 7 & 9 - 9 - 0 \\ -3 + 3 - 0 & 1 + 6 - 7 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{Here, } A^2 - 3A - 7I = 0$$

$$\Rightarrow A^{-1}[A^2 - 3A - 7I] = A^{-1}(0)$$

$$\Rightarrow A^{-1}A.A - 3A^{-1}A - 7A^{-1}I = 0$$

$$\Rightarrow IA - 3I - 7A^{-1} = 0$$

$$\Rightarrow -7A^{-1} = -A + 3I$$

$$\Rightarrow A^{-1} = \frac{-1}{7} \begin{bmatrix} -2 & -3 \\ 1 & 5 \end{bmatrix}$$

**12. Find the matrix A satisfying the matrix equation**

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

**Ans: Given:** Matrix equation.

Product of matrix is possible if number of rows of matrix is equal to number of columns of another matrix.

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}_{2 \times 2} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$$

$$\text{Let, } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2}$$

$$\Rightarrow \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2} \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$$

$$\Rightarrow \begin{bmatrix} 2a + c & 2b + d \\ 3a + 2c & 3b + 2d \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -6a - 3c + 10b + 5d & 4a + 2c - 6b - 3d \\ -9a - 6c + 15b + 10d & 6a + 4c - 9b - 6d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow -6a - 3c + 10b + 5d = 1 \quad \text{.....(i)}$$

$$\Rightarrow 4a + 2c - 6b - 3d = 0 \quad \text{.....(ii)}$$

$$\Rightarrow -9a - 6c + 15b + 10d = 0 \quad \text{.....(iii)}$$

$$\Rightarrow 6a + 4c - 9b - 6d = 1 \quad \text{.....(iv)}$$

add equation (i) and (iv),

$$c + b - d = 2$$

$$\Rightarrow d = c + b - 2 \quad \text{.....(v)}$$

add (ii) and (iii), we get

$$-5a - 4c + 9b + 7d = 0$$

add (vi) and (iv),

$$\Rightarrow d = 1 - a$$

from equation (v) and (vii), we get

$$\Rightarrow a = 3 - b - c \quad \dots(\text{viii})$$

use a and d in equation (iii), we get

$$-9(3 - b - c) - 6c + 15b + 10(-2 + b + c) = 0$$

$$\Rightarrow -27 + 9b + 9c - 6c + 15b - 20 + 10b + 10c = 0$$

$$\Rightarrow 34b + 13c = 47 \quad \dots(\text{ix})$$

use a and d in eq(ii), we get

$$4(3 - b - c) + 2c - 6b - 3(b + c - 2) = 0$$

$$\Rightarrow 12 - 4b - 4c + 2c - 6b - 3b - 3c + 6 = 0$$

$$\Rightarrow -13b - 5c = -18 \quad \dots(\text{x})$$

Solving equation (ix) and (x), we get

$$b = 1 \text{ and } c = 1$$

$$\text{and } a = 3 - 1 - 1 = 1, d = 1 - 1 = 0$$

$$\therefore A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

13. Find A, if  $\begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} A = \begin{bmatrix} -4 & 8 & 4 \\ -1 & 2 & 1 \\ -3 & 6 & 3 \end{bmatrix}$ .

**Ans: Given:** Matrix equation.

First, find the order of matrix using matrix multiplication.

Let  $A = \begin{bmatrix} x & y & z \end{bmatrix}_{1 \times 3}$

$$\begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}_{3 \times 1} \begin{bmatrix} x & y & z \end{bmatrix}_{1 \times 3} = \begin{bmatrix} -4 & 8 & 4 \\ -1 & 2 & 1 \\ -3 & 6 & 3 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow \begin{bmatrix} 4x & 4y & 4z \\ x & y & z \\ 3x & 3y & 3z \end{bmatrix} = \begin{bmatrix} -4 & 8 & 4 \\ -1 & 2 & 1 \\ -3 & 6 & 3 \end{bmatrix}$$

$$\Rightarrow 4x = -4, x = -1$$

$$\Rightarrow 4y = 8, y = 2$$

$$\Rightarrow 4z = 4, z = 1$$

$$\therefore A = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$$

14. If  $A = \begin{bmatrix} 3 & -4 \\ 1 & 1 \\ 2 & 0 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$ , then verify  $(BA)^2 = B^2A^2$ .

**Ans: Given:** Matrices A and B.

Substitute the matrix A and B in the equation. Use matrix multiplication.

Here,

$$BA = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 3 & -4 \\ 1 & 1 \\ 2 & 0 \end{bmatrix}_{3 \times 2}$$

$$BA = \begin{bmatrix} 11 & -7 \\ 13 & -2 \end{bmatrix}$$

$$\text{now, } (BA).(BA) = \begin{bmatrix} 11 & -7 \\ 13 & -2 \end{bmatrix} \cdot \begin{bmatrix} 11 & -7 \\ 13 & -2 \end{bmatrix}$$

$$\Rightarrow (BA)^2 = \begin{bmatrix} 30 & -63 \\ 117 & -87 \end{bmatrix}$$

As,  $B^2$  is not possible.

Therefore,  $(BA)^2 \neq B^2A^2$ .

**15. If possible, find the value of BA and AB, where**

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 1 & 2 \end{bmatrix}.$$

**Ans: Given:** Matrices A and B.

First, find BA and AB is possible or not. Then, use matrix multiplication.

Both BA and AB is possible.

Since in both B.A and A.B, the number of rows in one matrix is equal to number of columns in other matrix.

$$AB = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 1 & 2 \end{bmatrix}_{3 \times 2}$$

$$\Rightarrow AB = \begin{bmatrix} 12 & 9 \\ 12 & 15 \end{bmatrix}$$

$$\text{and } BA = \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 1 & 2 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}_{2 \times 3}$$

$$\Rightarrow BA = \begin{bmatrix} 9 & 6 & 12 \\ 7 & 8 & 16 \\ 4 & 5 & 10 \end{bmatrix}.$$

**16. Show by an example that for  $A \neq 0, B \neq 0$  and  $AB = 0$ .**

**Ans: Given:**  $A \neq 0, B \neq 0$ .

Consider matrix  $A$  and  $B$  such that  $A \neq 0, B \neq 0$ .

Let us consider that,

$$A = \begin{bmatrix} 0 & -4 \\ 0 & 2 \end{bmatrix} \neq 0 \text{ and } B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix} \neq 0$$

$$\therefore AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0.$$

**17. Given,  $A = \begin{bmatrix} 2 & 4 & 0 \\ 3 & 9 & 6 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 4 \\ 2 & 8 \\ 1 & 3 \end{bmatrix}$ . Is  $(AB)' = B'A'$ ?**

**Ans: Given:** Matrix  $A$  and  $B$ .

Substitute the matrix  $A$  and  $B$  in the equation. Use matrix multiplication.

$$AB = \begin{bmatrix} 2 & 4 & 0 \\ 3 & 9 & 6 \end{bmatrix} \cdot \begin{bmatrix} 1 & 4 \\ 2 & 8 \\ 1 & 3 \end{bmatrix}$$

$$AB = \begin{bmatrix} 10 & 40 \\ 27 & 102 \end{bmatrix}$$

$$(AB)' = \begin{bmatrix} 10 & 27 \\ 40 & 102 \end{bmatrix} \quad \dots(i)$$

$$\text{now, } B' = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 8 & 3 \end{bmatrix} \text{ and } A' = \begin{bmatrix} 2 & 3 \\ 4 & 9 \\ 0 & 6 \end{bmatrix}$$

$$B'A' = \begin{bmatrix} 10 & 27 \\ 40 & 102 \end{bmatrix} \quad \dots(ii)$$

from equation (i) and (ii), we get

$$\therefore (AB)' = B'A'$$

**18. Solve for x and y,  $x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 3 \\ 5 \end{bmatrix} + \begin{bmatrix} -8 \\ -11 \end{bmatrix} = 0$ .**

**Ans: Given:** Matrix equation.

Use matrix algebra to find x and y.

Here,

$$x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 3 \\ 5 \end{bmatrix} + \begin{bmatrix} -8 \\ -11 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 2x \\ x \end{bmatrix} + \begin{bmatrix} 3y \\ 5y \end{bmatrix} + \begin{bmatrix} -8 \\ -11 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2x + 3y - 8 \\ x + 5y - 11 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow 2x + 3y = 8 \text{ .....(i)}$$

$$\Rightarrow x + 5y = 11 \text{ .....(ii)}$$

from equation (i) and (ii), we get

$x = 1$  and  $y = 2$ .

**19. If X and Y are  $2 \times 2$  matrices, then solve the following matrix equations for X and Y,**

$$2X + 3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix}, 3X + 2Y = \begin{bmatrix} -2 & 2 \\ 1 & -5 \end{bmatrix}.$$

**Ans: Given:** X and Y are  $2 \times 2$  matrices.

Solve the equation using matrix algebra.

Here,

$$2X + 3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \text{ .....(i)}$$

$$3X + 2Y = \begin{bmatrix} -2 & 2 \\ 1 & -5 \end{bmatrix} \text{ .....(ii)}$$

Subtract equation (i) from (ii), we get

$$(3X + 2Y) - (2X + 3Y) = \begin{bmatrix} -2 - 2 & 2 - 3 \\ 1 - 4 & -5 - 0 \end{bmatrix}$$

$$X - Y = \begin{bmatrix} -4 & -1 \\ -3 & -5 \end{bmatrix} \quad \dots(iii)$$

now add (i) and (ii),

$$5X + 5Y = \begin{bmatrix} 0 & 5 \\ 5 & -5 \end{bmatrix}$$

$$\Rightarrow X + Y = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \quad \dots(iv)$$

now add (iii) and (iv),

$$(X - Y) + (X + Y) = \begin{bmatrix} -4 & 0 \\ 2 & -6 \end{bmatrix}$$

$$\Rightarrow X = \begin{bmatrix} -2 & 0 \\ -1 & -3 \end{bmatrix}$$

substituting in equation (iv), we get

$$\begin{bmatrix} -2 & 0 \\ -1 & -3 \end{bmatrix} + Y = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\Rightarrow Y = \begin{bmatrix} 2 & 1 \\ 2 & 2 \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} -2 & 0 \\ -1 & -3 \end{bmatrix} \text{ and } Y = \begin{bmatrix} 2 & 1 \\ 2 & 2 \end{bmatrix}.$$

**20. If  $A = \begin{bmatrix} 3 & 5 \end{bmatrix}$  and  $B = \begin{bmatrix} 7 & 3 \end{bmatrix}$ , then find a non-zero matrix  $C$  such that  $AC = BC$ .**

**Ans: Given:**  $A = \begin{bmatrix} 3 & 5 \end{bmatrix}$  and  $B = \begin{bmatrix} 7 & 3 \end{bmatrix}$ .

Consider a matrix with defined order. Then, substitute in the given condition.

Here,

$$A = \begin{bmatrix} 3 & 5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & 3 \end{bmatrix}.$$

Let,

$$C = \begin{bmatrix} x \\ y \end{bmatrix}_{2 \times 1}$$

$$AC = \begin{bmatrix} 3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\therefore AC = \begin{bmatrix} 3x + 5y \end{bmatrix}$$

$$\text{now, } BC = \begin{bmatrix} 7 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\therefore BC = \begin{bmatrix} 7x + 3y \end{bmatrix}$$

Since,  $AC = BC$ ,

$$\Rightarrow \begin{bmatrix} 3x + 5y \end{bmatrix} = \begin{bmatrix} 7x + 3y \end{bmatrix}$$

$$\Rightarrow 3x + 5y = 7x + 3y$$

$$\Rightarrow x = \frac{1}{2}y$$

$$\Rightarrow y = 2x$$

$$\therefore C = \begin{bmatrix} x \\ 2x \end{bmatrix}$$

Considering the order  $2 \times 1, 2 \times 2, 2 \times 3, \dots$  of C. We get,

$$C = \begin{bmatrix} x \\ 2x \end{bmatrix} \begin{bmatrix} x & x \\ 2x & 2x \end{bmatrix} \dots$$

In general,

$$C = \begin{bmatrix} k \\ 2k \end{bmatrix} \begin{bmatrix} k & k \\ 2k & 2k \end{bmatrix} \dots$$

where, k is any real number.

**21. Give an example of matrices A, B and C, such that  $AB = AC$ , where A is non-zero matrix but  $B \neq C$ .**

**Ans: Given:** Matrices A, B and C.

Consider matrices A, B and C, such that  $AB = AC$  and  $B \neq C$ .

Let us consider the matrices, A, B and C.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \text{ and } C = \begin{bmatrix} 2 & 3 \\ 4 & 3 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 2 & 3 \\ 0 & 0 \end{bmatrix}$$

now,

$$AC = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 3 \end{bmatrix}$$

$$\Rightarrow AC = \begin{bmatrix} 2 & 3 \\ 0 & 0 \end{bmatrix}$$

$\therefore AB = AC$ .

22. If  $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & 3 \\ 3 & -4 \end{bmatrix}$  and  $C = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$ , verify

(i)  $(AB)C = A(BC)$ .

**Ans: Given:** Matrices A, B and C.

Consider left and right hand side of equation. Then, substitute the matrices and verify.

Here,

$$AB = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & -4 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 8 & -5 \\ -1 & -10 \end{bmatrix}$$

$$(AB)C = \begin{bmatrix} 8 & -5 \\ -1 & -10 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$$

$$\therefore (AB)C = \begin{bmatrix} 13 & 0 \\ 9 & 0 \end{bmatrix} \quad \dots(i)$$

Now,

$$(BC) = \begin{bmatrix} 2 & 3 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$$

$$\Rightarrow BC = \begin{bmatrix} -1 & 0 \\ 7 & 0 \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 7 & 0 \end{bmatrix}$$

$$\Rightarrow A(BC) = \begin{bmatrix} 13 & 0 \\ 9 & 0 \end{bmatrix}$$

(ii)  $A(B + C) = AB + AC$ .

**Ans: Given:** Matrices A, B and C.

Consider left and right hand side of equation. Then, substitute the matrices and verify.

Here,

$$(B + C) = \begin{bmatrix} 2 & 3 \\ 3 & -4 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$$

$$\Rightarrow (B + C) = \begin{bmatrix} 3 & 3 \\ 2 & -4 \end{bmatrix}$$

$$A.(B + C) = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 2 & -4 \end{bmatrix}$$

$$\Rightarrow A.(B + C) = \begin{bmatrix} 7 & -5 \\ -4 & -10 \end{bmatrix}$$

now,

$$AB = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & -4 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 8 & -5 \\ -1 & -10 \end{bmatrix}$$

$$AC = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$$

$$\Rightarrow AC = \begin{bmatrix} -1 & 0 \\ -3 & 0 \end{bmatrix}$$

$$AB + AC = \begin{bmatrix} 8 & -5 \\ -1 & -10 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ -3 & 0 \end{bmatrix}$$

$$\therefore AB + AC = \begin{bmatrix} 7 & -5 \\ -4 & -10 \end{bmatrix} \dots\dots(ii)$$

from (i) and (ii),

$$A.(B + C) = AB + AC$$

23. If  $P = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$  and  $Q = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ , then prove that

$$PQ = \begin{bmatrix} xa & 0 & 0 \\ 0 & yb & 0 \\ 0 & 0 & zc \end{bmatrix} = QP.$$

**Ans: Given:** Matrices P and Q.

Use matrix multiplication. Then, compare P and Q.

$$PQ = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$\Rightarrow PQ = \begin{bmatrix} xa & 0 & 0 \\ 0 & yb & 0 \\ 0 & 0 & zc \end{bmatrix} \dots\dots(i)$$

$$QP = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$$

$$\Rightarrow QP = \begin{bmatrix} xa & 0 & 0 \\ 0 & yb & 0 \\ 0 & 0 & zc \end{bmatrix} \dots(ii)$$

from equation (i) and (ii),

$$PQ = QP.$$

24. If  $\begin{bmatrix} 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = A$ , then find value of A.

**Ans: Given:** Matrix equation.

Solve the matrix equation using matrix multiplication.

$$\begin{bmatrix} 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = A$$

$$\Rightarrow \begin{bmatrix} -3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = A$$

$$\Rightarrow A = \begin{bmatrix} -4 \end{bmatrix}$$

25. If  $A = \begin{bmatrix} 2 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix}$  and  $C = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ , then verify that

$$A(B + C) = (AB + AC).$$

**Ans: Given:** Matrices A, B and C.

Substitute the matrices in left and right hand side of matrix equation. Then, compare.

$$B + C = \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix} + \begin{bmatrix} -1 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$\therefore B + C = \begin{bmatrix} 4 & 5 & 5 \\ 9 & 7 & 8 \end{bmatrix}$$

$$A(B + C) = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 5 & 5 \\ 9 & 7 & 8 \end{bmatrix} \quad \dots(i)$$

$$\therefore A(B + C) = \begin{bmatrix} 17 & 17 & 18 \end{bmatrix}$$

now,

$$AB = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 18 & 13 & 14 \end{bmatrix}$$

$$AC = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$\Rightarrow AC = \begin{bmatrix} -1 & 4 & 4 \end{bmatrix}$$

$$AB + AC = \begin{bmatrix} 17 & 17 & 18 \end{bmatrix} \quad \dots(ii)$$

from equation (i) and (ii), we get

$$\therefore A(B + C) = AB + AC$$

26. If  $A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix}$ , then verify that  $A^2 + A = (A + I)$ , where I is  $3 \times 3$  unit matrix.

**Ans: Given:** Matrix A.

First, find  $A^2$ . Then, substitute the values in the matrix equation.

$$A^2 = A.A$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & -1 & -2 \\ 4 & 4 & 4 \\ 2 & 2 & 4 \end{bmatrix}$$

$$\Rightarrow A^2 + A = \begin{bmatrix} 2 & -1 & -3 \\ 6 & 5 & 7 \\ 2 & 3 & 5 \end{bmatrix} \quad \text{.....(i)}$$

now,

$$A(A + I) = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix} \left( \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

$$\Rightarrow A(A + I) = \begin{bmatrix} 2 & -1 & -3 \\ 6 & 5 & 7 \\ 2 & 3 & 5 \end{bmatrix} \quad \text{.....(ii)}$$

from equation (i) and (ii), we get

$$A^2 + A = A(A + I)$$

27. If  $A = \begin{bmatrix} 0 & -1 & 2 \\ 4 & 3 & -4 \end{bmatrix}$  and  $B = \begin{bmatrix} 4 & 0 \\ 1 & 3 \\ 2 & 6 \end{bmatrix}$ , then verify that

(i)  $(A')' = A$

**Ans: Given:** Matrix A.

Substitute the matrix in the given equations.

Here,

$$A' = \begin{bmatrix} 0 & 4 \\ -1 & 3 \\ 2 & -4 \end{bmatrix}$$

$$\Rightarrow (A')' = \begin{bmatrix} 0 & -1 & 2 \\ 4 & 3 & -4 \end{bmatrix}$$

$$\therefore (A')' = A$$

**(ii)  $(AB)' = B'A'$**

**Ans: Given:** Matrix A.

Substitute the matrix in the given equations.

Here,

$$AB = \begin{bmatrix} 3 & 9 \\ 11 & -15 \end{bmatrix}$$

$$\Rightarrow (AB)' = \begin{bmatrix} 3 & 11 \\ 9 & -15 \end{bmatrix}$$

$$B'A' = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & 6 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ -1 & 3 \\ 2 & -4 \end{bmatrix}$$

$$\Rightarrow B'A' = \begin{bmatrix} 3 & 11 \\ 9 & -15 \end{bmatrix}$$

$$\therefore B'A' = (AB)'$$

**(iii)  $(kA)' = (kA')$ .**

**Ans: Given:** Matrix A.

Substitute the matrix in the given equations.

Here,

$$(kA) = \begin{bmatrix} 0 & -k & 2k \\ 4k & 3k & -4k \end{bmatrix}$$

$$\Rightarrow (kA)' = \begin{bmatrix} 0 & 4k \\ -k & 3k \\ 2k & -4k \end{bmatrix}$$

$$kA' = \begin{bmatrix} 0 & 4k \\ -k & 3k \\ 2k & -4k \end{bmatrix}$$

$$\Rightarrow kA' = (kA)'$$

28. If  $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 5 & 6 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 2 \\ 6 & 4 \\ 7 & 3 \end{bmatrix}$ , then verify that

(i)  $(2A + B)' = 2A' + B'$

**Ans: Given:** Matrices A and B.

Substitute the matrices A and B. Then, use matrix algebra.

Here,

$$(2A + B) = \begin{bmatrix} 2 & 4 \\ 8 & 2 \\ 10 & 12 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 6 & 4 \\ 7 & 3 \end{bmatrix}$$

$$\Rightarrow 2A + B = \begin{bmatrix} 3 & 6 \\ 14 & 6 \\ 17 & 15 \end{bmatrix}$$

$$\Rightarrow (2A + B)' = \begin{bmatrix} 3 & 14 & 17 \\ 6 & 6 & 15 \end{bmatrix}$$

now,

$$2A' + B' = 2 \begin{bmatrix} 1 & 4 & 5 \\ 2 & 1 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 6 & 7 \\ 2 & 4 & 3 \end{bmatrix}$$

$$\Rightarrow 2A' + B' = \begin{bmatrix} 3 & 14 & 17 \\ 6 & 6 & 15 \end{bmatrix}$$

$$\therefore 2A' + B' = (2A + B)'$$

$$\text{(ii) } (A - B)' = A' - B'.$$

**Ans: Given:** Matrices A and B.

Substitute the matrices A and B. Then, use matrix algebra.

Here,

$$A - B = \begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 5 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 6 & 4 \\ 7 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -2 & -3 \\ -2 & 3 \end{bmatrix}$$

$$\Rightarrow (A - B)' = \begin{bmatrix} 0 & -2 & -2 \\ 0 & -3 & 3 \end{bmatrix}$$

now,

$$A' - B' = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 1 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 6 & 7 \\ 2 & 4 & 3 \end{bmatrix}$$

$$\Rightarrow A' - B' = \begin{bmatrix} 0 & -2 & -2 \\ 0 & -3 & 3 \end{bmatrix}$$

$$\therefore A' - B' = (A - B)'$$

**29. Show that  $A'A$  and  $AA'$  are both symmetric matrices for any matrix A.**

**Ans: Given:** Matrix A.

For symmetric matrix  $A' = A$  and for skew symmetric matrix  $A' = -A$ .

Let

$$X = A'A$$

$$\Rightarrow X' = (A'A)'$$

$$\Rightarrow X' = A'(A)'$$

$$\Rightarrow X' = A'A$$

$$\Rightarrow X' = X$$

$\therefore A'A$  is a symmetric matrix

$$Y = AA'$$

$$\Rightarrow Y' = (AA)'$$

$$\Rightarrow Y' = (A')A'$$

$$\Rightarrow Y' = AA'$$

$$\Rightarrow Y' = Y$$

$\therefore AA'$  is symmetric matrix.

**30. Let A and B be square matrices of order  $3 \times 3$ . Is  $(AB)^2 = A^2B^2$ ? Give reasons.**

**Ans: Given:** A and B are square matrices.

Use the matrix multiplication. Then find the condition for which  $(AB)^2 = A^2B^2$ .

Here, A and B are square matrices.

$$(AB)^2 = AB.AB$$

$$\Rightarrow (AB)^2 = AABB$$

$$\Rightarrow (AB)^2 = A^2B^2$$

$\therefore (AB)^2 = A^2B^2$  is true, when  $AB = BA$ .

**31. Show that, if A and B are square matrices such that  $AB = BA$ , then  $(A + B)^2 = A^2 + 2AB + B^2$ .**

**Ans: Given:** A and B and  $AB = BA$ .

Use  $(A + B)^2 = (A + B)(A + B)$ . Then, use matrix algebra.

Here,

$$AB = BA$$

$$(A + B)^2 = (A + B)(A + B)$$

$$\Rightarrow (A + B)^2 = A^2 + AB + BA + B^2$$

$$\Rightarrow (A + B)^2 = A^2 + AB + AB + B^2$$

$$\therefore (A + B)^2 = A^2 + 2AB + B^2$$

**32. If  $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ ,  $B = \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix}$ ,  $C = \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix}$ , a = 4 and b = -2, then show that**

**(i)  $A + (B + C) = (A + B) + C$**

**Ans: Given:** Matrices A, B and C.

Substitute the matrices A, B and C in the matrix equation.

Here,

$$A + (B + C) = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$\Rightarrow A + (B + C) = \begin{bmatrix} 7 & 2 \\ 1 & 6 \end{bmatrix}$$

$$(A + B) + C = \begin{bmatrix} 5 & 2 \\ 0 & 8 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix}$$

$$\Rightarrow (A + B) + C = \begin{bmatrix} 7 & 2 \\ 1 & 6 \end{bmatrix}$$

$$\therefore A + (B + C) = (A + B) + C$$

**(ii)  $A(BC) = (AB)C$**

**Ans: Given:** Matrices A, B and C.

Substitute the matrices A, B and C in the matrix equation.

Here,

$$BC = \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix}$$

$$\Rightarrow BC = \begin{bmatrix} 8 & 0 \\ 7 & -10 \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 7 & -10 \end{bmatrix}$$

$$\Rightarrow A(BC) = \begin{bmatrix} 22 & -20 \\ 13 & -30 \end{bmatrix}$$

now,

$$(AB) = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix}$$

$$\Rightarrow (AB) = \begin{bmatrix} 6 & 10 \\ -1 & 15 \end{bmatrix}$$

$$(AB)C = \begin{bmatrix} 6 & 10 \\ -1 & 15 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix}$$

$$\Rightarrow (AB)C = \begin{bmatrix} 22 & -20 \\ 13 & -30 \end{bmatrix}$$

**(iii)  $(a + b)B = aB + bB$**

**Ans: Given:** Matrices A, B and C.

Substitute the matrices A, B and C in the matrix equation.

Here,

$$(a + b)B = (4 - 2) \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix}$$

$$\Rightarrow (a + b)B = \begin{bmatrix} 8 & 0 \\ 2 & 10 \end{bmatrix}$$

$$aB + bB = 4B - 2B$$

$$\Rightarrow aB + bB = \begin{bmatrix} 16 & 0 \\ 4 & 20 \end{bmatrix} - \begin{bmatrix} 8 & 0 \\ 2 & 10 \end{bmatrix}$$

$$\Rightarrow aB + bB = \begin{bmatrix} 8 & 0 \\ 2 & 10 \end{bmatrix}$$

$$\Rightarrow aB + bB = (a + b)B$$

**(iv)  $a(C - A) = aC - aA$**

**Ans: Given:** Matrices A, B and C.

Substitute the matrices A, B and C in the matrix equation.

Here,

$$(C - A) = \begin{bmatrix} 1 & -2 \\ 2 & -5 \end{bmatrix}$$

$$\Rightarrow a(C - A) = \begin{bmatrix} 4 & -8 \\ 8 & -20 \end{bmatrix}$$

$$aC - aA = \begin{bmatrix} 8 & 0 \\ 4 & -8 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ -4 & 12 \end{bmatrix}$$

$$\Rightarrow aC - aA = \begin{bmatrix} 4 & -8 \\ 8 & -20 \end{bmatrix}$$

$$\Rightarrow aC - aA = a(C - A)$$

**(v)  $(A^T)^T = A$**

**Ans: Given:** Matrices A, B and C.

Substitute the matrices A, B and C in the matrix equation.

Here,

$$A^T = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}^T = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$$

$$(A^T)^T = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}^T = A$$

**(vi)  $(bA)^T = bA^T$**

**Ans: Given:** Matrices A, B and C.

Substitute the matrices A, B and C in the matrix equation.

Here,

$$(bA)^T = \begin{bmatrix} -2 & -4 \\ 2 & -6 \end{bmatrix}^T = \begin{bmatrix} -2 & 2 \\ -4 & -6 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$$

$$\Rightarrow bA^T = \begin{bmatrix} -2 & 2 \\ -4 & -6 \end{bmatrix}$$

$$\Rightarrow bA^T = (bA)^T$$

**(vii)  $(AB)^T = B^T A^T$**

**Ans: Given:** Matrices A, B and C.

Substitute the matrices A, B and C in the matrix equation.

Here,

$$AB = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 10 \\ -1 & 15 \end{bmatrix}$$

$$\Rightarrow (AB)^T = \begin{bmatrix} 6 & -1 \\ 10 & 15 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} 4 & 1 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 6 & -1 \\ 10 & 15 \end{bmatrix}$$

$$\Rightarrow B^T A^T = (AB)^T$$

**(viii)  $(A - B)C = AC - BC$**

**Ans: Given:** Matrices A, B and C.

Substitute the matrices A, B and C in the matrix equation.

Here,

$$(A - B) = \begin{bmatrix} -3 & 2 \\ -2 & -2 \end{bmatrix}$$

$$\Rightarrow (A - B)C = \begin{bmatrix} -4 & -4 \\ -6 & 4 \end{bmatrix}$$

$$AC = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ 1 & -6 \end{bmatrix}$$

$$BC = \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 7 & -10 \end{bmatrix}$$

$$\Rightarrow AC - BC = \begin{bmatrix} -4 & -4 \\ -6 & 4 \end{bmatrix}$$

$$\Rightarrow AC - BC = (A - B)C$$

$$\text{(ix)} (A - B)^T = A^T - B^T$$

**Ans: Given:** Matrices A, B and C.

Substitute the matrices A, B and C in the matrix equation.

Here,

$$(A - B)^T = \begin{bmatrix} -3 & 2 \\ -2 & -2 \end{bmatrix}^T = \begin{bmatrix} -3 & -2 \\ 2 & -2 \end{bmatrix}$$

$$A^T - B^T = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 1 \\ 0 & 5 \end{bmatrix}$$

$$\Rightarrow A^T - B^T = \begin{bmatrix} -3 & -2 \\ 2 & -2 \end{bmatrix}$$

$$\Rightarrow A^T - B^T = (A - B)^T$$

33. If  $A = \begin{bmatrix} \cos q & \sin q \\ -\sin q & \cos q \end{bmatrix}$ , then show that  $A^2 = \begin{bmatrix} \cos 2q & \sin 2q \\ -\sin 2q & \cos 2q \end{bmatrix}$ .

**Ans: Given:** Matrix A.

Use  $A^2 = A.A$ . Then, substitute matrix A.

$$A^2 = A.A$$

$$\Rightarrow A^2 = \begin{bmatrix} \cos 2q & \sin 2q \\ -\sin 2q & \cos 2q \end{bmatrix} \begin{bmatrix} \cos 2q & \sin 2q \\ -\sin 2q & \cos 2q \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} \cos^2 q - \sin^2 q & \cos q \sin q + \sin q \cos q \\ -\cos q \sin q - \sin q \cos q & -\sin^2 q + \cos^2 q \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} \cos 2q & 2\sin q \cos q \\ -2\sin q \cos q & \cos 2q \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} \cos 2q & \sin 2q \\ -\sin 2q & \cos 2q \end{bmatrix}.$$

34. If  $A = \begin{bmatrix} 0 & -x \\ x & 0 \end{bmatrix}$ ,  $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  and  $x^2 = -1$ , then show that  $(A + B)^2 = A^2 + B^2$ .

**Ans: Given:** Matrices A, B and  $x^2 = -1$ .

Substitute the matrices in equation. Use matrix algebra.

$$(A + B) = \begin{bmatrix} 0 & -x + 1 \\ x + 1 & 0 \end{bmatrix}$$

$$\Rightarrow (A + B)^2 = \begin{bmatrix} 0 & -x + 1 \\ x + 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -x + 1 \\ x + 1 & 0 \end{bmatrix}$$

$$\Rightarrow (A + B)^2 = \begin{bmatrix} 1 - x^2 & 0 \\ 0 & 1 - x^2 \end{bmatrix}$$

now,

$$A^2 = A.A$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & -x \\ x & 0 \end{bmatrix} \begin{bmatrix} 0 & -x \\ x & 0 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} -x^2 & 0 \\ 0 & x^2 \end{bmatrix}$$

$$B^2 = B.B$$

$$\Rightarrow B^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow B^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 + B^2 = \begin{bmatrix} 1 - x^2 & 0 \\ 0 & 1 - x^2 \end{bmatrix}$$

$$\Rightarrow (A + B)^2 = A^2 + B^2$$

35. Verify  $A^2 = I$ , where  $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ .

**Ans: Given:** Matrix A.

First, find  $A^2$  using  $A^2 = A.A$ . Then, compare it with identity matrix.

$$A^2 = A.A$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore A^2 = I.$$

**36. Prove by mathematical induction that  $(A')^n = (A^n)'$  where  $n \in \mathbb{N}$  for any square matrix  $A$ .**

**Ans: Given:** Square matrix  $A$ .

Consider  $P(1)$  and  $P(k)$  be true. Then, prove that  $P(k + 1)$  be true.

Let,

$$P(n) : (A')^n = (A^n)' \text{ be true.}$$

$$P(1) : A' = A$$

$$\Rightarrow P(1) \text{ be true.}$$

now,  $P(k)$  be true,

$$P(k) : (A')^k = (A^k)'$$

now,  $P(k + 1)$

$$P(k + 1) : (A')^{k+1} = (A^{k+1})'$$

$$P(k + 1) : (A')^k \cdot (A')^1 = [A^{k+1}]'$$

$$P(k + 1) : (A^k)' \cdot (A)' = (A')^{k+1} = [A^{k+1}]'$$

$$P(k+1) : (A.A^k)' = [A^{k+1}]'$$

$$P(k+1) : [A^{k+1}]' = [A^{k+1}]'$$

$\therefore P(k+1)$  is true.

**37. Find inverse, by elementary row operations (if possible), of the following matrices.**

$$(i) \begin{bmatrix} 1 & 3 \\ -5 & 7 \end{bmatrix}$$

**Ans: Given:** Matrices.

For elementary row operations use  $A = IA$ .

Here,

$$A = \begin{bmatrix} 1 & 3 \\ -5 & 7 \end{bmatrix}$$

For elementary row operation,

$$A = IA.$$

$$\Rightarrow \begin{bmatrix} 1 & 3 \\ -5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$$

$$\Rightarrow \begin{bmatrix} 1 & 3 \\ 0 & 22 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} A \quad [\text{operation: } R_2 \rightarrow R_2 + 5R_1]$$

$$\Rightarrow \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{5}{22} & \frac{1}{22} \end{bmatrix} A \quad \left[ \text{operation: } R_2 \rightarrow \frac{1}{22} R_2 \right]$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{7}{22} & \frac{-3}{22} \\ \frac{5}{22} & \frac{1}{22} \end{bmatrix} A \quad [\text{operation: } R_1 \rightarrow R_1 - 3R_2]$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{22} \begin{bmatrix} 7 & -3 \\ 5 & 1 \end{bmatrix} A$$

$$\Rightarrow I = A^{-1}A$$

$$\therefore A^{-1} = \frac{1}{22} \begin{bmatrix} 7 & -3 \\ 5 & 1 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 1 & -3 \\ -2 & 6 \end{bmatrix}$$

**Ans: Given:** Matrices.

For elementary row operations use  $A = IA$ .

Here,

$$A = \begin{bmatrix} 1 & -3 \\ -2 & 6 \end{bmatrix}$$

For elementary row operation,

$$A = IA$$

$$\Rightarrow \begin{bmatrix} 1 & -3 \\ -2 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$$

$$\Rightarrow \begin{bmatrix} 1 & -3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} A \quad [\text{operation: } R_2 \rightarrow R_2 + 2R_1]$$

All zeros in a row represent that  $A^{-1}$  does not exist.

38. If  $\begin{bmatrix} xy & 4 \\ z+6 & x+y \end{bmatrix} = \begin{bmatrix} 8 & w \\ 0 & 6 \end{bmatrix}$ , then find the values of  $x, y, z$  and  $w$ .

**Ans: Given:** Matrix equation.

If two matrix are equal, then  $a_{ij} = b_{ij}$ .

Here,

$$\begin{bmatrix} xy & 4 \\ z+6 & x+y \end{bmatrix} = \begin{bmatrix} 8 & w \\ 0 & 6 \end{bmatrix}$$

$$\Rightarrow x+y=6 \text{ and } xy=8$$

$$\Rightarrow (6-y)y=8$$

$$\Rightarrow y^2 - 6y + 8 = 0$$

$$\Rightarrow y^2 - 4y - 2y + 8 = 0$$

$$\Rightarrow (y-2)(y-4) = 0$$

$$\Rightarrow y=2 \text{ or } y=4$$

$$\Rightarrow x=6-2=4$$

$$\Rightarrow x=6-4=2$$

$$\text{now, } z+6=0 \text{ and } w=4$$

$$\therefore x=2, y=4 \text{ or } x=4, y=2, z=-6 \text{ and } w=4.$$

39. If  $A = \begin{bmatrix} 1 & 5 \\ 7 & 12 \end{bmatrix}$  and  $B = \begin{bmatrix} 9 & 1 \\ 7 & 8 \end{bmatrix}$ , then find a matrix  $C$  such that

$3A + 5B + 2C$  is a null matrix.

**Ans: Given:** Matrices  $A$  and  $B$ .

Substitute the matrices  $A$  and  $B$ . Then, use matrix algebra.

Here,

Let,

$$C = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$3A + 5B + 2C = 0$$

$$\Rightarrow \begin{bmatrix} 3 & 15 \\ 21 & 36 \end{bmatrix} + \begin{bmatrix} 45 & 5 \\ 35 & 40 \end{bmatrix} + \begin{bmatrix} 2a & 2b \\ 2c & 2d \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 48 + 2a & 20 + 2b \\ 56 + 2c & 76 + 2d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow 2a + 48 = 0 \text{ or } a = -24$$

$$\Rightarrow 20 + 2b = 0 \text{ or } b = -10$$

$$\Rightarrow 56 + 2c = 0 \text{ or } c = -28$$

$$\Rightarrow 76 + 2d = 0 \text{ or } d = -38$$

$$\therefore C = \begin{bmatrix} -24 & -10 \\ -28 & -38 \end{bmatrix}$$

**40. If  $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$ , then  $A^2 - 5A - 14I$ . Hence, obtain  $A^3$ .**

**Ans: Given:** Matrix  $A$ .

First, calculate  $A^2$  using  $A^2 = A.A$  and then  $A^3$ .

$$A^2 = A.A$$

$$\Rightarrow A^2 = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix}$$

$$\therefore A^2 - 5A - 14I = \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix} - \begin{bmatrix} 15 & -25 \\ -20 & 10 \end{bmatrix} - \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix}$$

$$\Rightarrow A^2 - 5A - 14I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow A^2 - 5A - 14I = 0$$

$$\Rightarrow A^3 - 5A^2 - 14A = 0$$

$$\Rightarrow A^3 = 5A^2 + 14A$$

$$\Rightarrow A^3 = \begin{bmatrix} 145 & -125 \\ -100 & 120 \end{bmatrix} + \begin{bmatrix} 42 & -70 \\ -56 & 28 \end{bmatrix}$$

$$\therefore A^3 = \begin{bmatrix} 187 & -195 \\ -156 & 148 \end{bmatrix}$$

**41. Find the values of a, b, c and d, if**

$$3 \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 6 \\ -1 & 2d \end{bmatrix} + \begin{bmatrix} 4 & a+b \\ c+d & 3 \end{bmatrix}$$

**Ans: Given:** Matrix equations.

Solve the equation using matrix algebra.

$$3 \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 6 \\ -1 & 2d \end{bmatrix} + \begin{bmatrix} 4 & a+b \\ c+d & 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3a & 3b \\ 3c & 3d \end{bmatrix} = \begin{bmatrix} a+4 & 6+a+b \\ c+d-1 & 2d+3 \end{bmatrix}$$

$$\Rightarrow 3a = a + 4 \text{ or } a = 2$$

$$\Rightarrow 3b = 6 + a + b \text{ or } b = 4$$

$$\Rightarrow 2c = c + d - 1 \text{ or } c = 1$$

$$\therefore a = 2, b = 4, c = 1 \text{ and } d = 3$$

42. Find the matrix A such that  $\begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{bmatrix} A = \begin{bmatrix} -1 & -8 & -10 \\ 1 & -2 & -5 \\ 9 & 22 & 15 \end{bmatrix}$ .

**Ans: Given:** Matrix equation.

Consider the matrix with defined order. Then, use matrix multiplication.

Here,

$$\begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{bmatrix}_{3 \times 2} A = \begin{bmatrix} -1 & -8 & -10 \\ 1 & -2 & -5 \\ 9 & 22 & 15 \end{bmatrix}_{3 \times 3}$$

Therefore, the possible order of A is  $2 \times 3$ .

$$\text{So, } A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = \begin{bmatrix} -1 & -8 & -10 \\ 1 & -2 & -5 \\ 9 & 22 & 15 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2a - d & 2b - e & 2c - f \\ a + 0.d & b + 0.e & c + 0.f \\ -3a + 4d & -3b + 4e & -3c + 4f \end{bmatrix} = \begin{bmatrix} -1 & -8 & -10 \\ 1 & -2 & -5 \\ 9 & 22 & 15 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2a - d & 2b - e & 2c - f \\ a & b & c \\ -3a + 4d & -3b + 4e & -3c + 4f \end{bmatrix} = \begin{bmatrix} -1 & -8 & -10 \\ 1 & -2 & -5 \\ 9 & 22 & 15 \end{bmatrix}$$

$$\Rightarrow a = 1, b = -2, c = -5$$

$$\text{and, } 2a - d = -1 \text{ or } d = 3$$

$$\Rightarrow 2b - e = -8 \text{ or } e = 4$$

$$\Rightarrow 2c - f = -10 \text{ or } f = 0$$

$$\therefore A = \begin{bmatrix} 1 & -2 & -5 \\ 3 & 4 & 0 \end{bmatrix}$$

**43: If  $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$ , then find  $A^2 + 2A + 7I$ .**

**Ans: Given:** Matrix A.

First, find  $A^2$  using  $A^2 = A.A$ . Then, substitute in the equation  $A^2 + 2A + 7I$ .

$$A^2 = A.A$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 9 & 4 \\ 8 & 9 \end{bmatrix}$$

$$\Rightarrow A^2 + 2A + 7I = \begin{bmatrix} 9 & 4 \\ 8 & 9 \end{bmatrix} + \begin{bmatrix} 2 & 4 \\ 8 & 2 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\therefore A^2 + 2A + 7I = \begin{bmatrix} 18 & 8 \\ 16 & 18 \end{bmatrix}$$

**44. If  $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$  and  $A^{-1} = A'$ , then find the value of  $\alpha$ .**

**Ans: Given:** Matrix A and  $A^{-1} = A'$ .

Use basic identity  $AA^{-1} = AA'$ .

$$A^{-1} = A'$$

$$\Rightarrow AA^{-1} = AA'$$

$$\Rightarrow I = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos^2\alpha + \sin^2\alpha & \sin\alpha \\ -\sin\alpha & \sin^2\alpha + \cos^2\alpha \end{bmatrix}$$

$$\Rightarrow \cos^2\alpha + \sin^2\alpha = 1$$

This is true for all real values of  $\alpha$ .

45. If matrix  $\begin{bmatrix} 0 & a & 3 \\ 2 & b & -1 \\ c & 1 & 0 \end{bmatrix}$  is a skew-symmetric matrix, then find the value of **a, b and c.**

**Ans: Given:** Skew symmetric matrix.

For skew symmetric matrix  $A' = -A$ .

$$A' = -A$$

$$\Rightarrow \begin{bmatrix} 0 & 2 & c \\ a & b & 1 \\ 3 & -1 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & a & 3 \\ 2 & b & -1 \\ c & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 2 & c \\ a & b & 1 \\ 3 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -a & -3 \\ -2 & -b & 1 \\ -c & -1 & 0 \end{bmatrix}$$

$$\Rightarrow a = -2, b = 0 \text{ and } c = -3.$$

46. If  $P(x) = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ , then show that  $P(x).P(y) = P(x+y) = P(y).P(x)$ .

**Ans: Given:** Matrix  $P(x)$ .

Replace  $x$  with  $y$  and  $(x+y)$  in matrix. Then, use matrix multiplication.

$$P(x) = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}, P(y) = \begin{bmatrix} \cos y & \sin y \\ -\sin y & \cos y \end{bmatrix}$$

$$\Rightarrow P(x).P(y) = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix} \begin{bmatrix} \cos y & \sin y \\ -\sin y & \cos y \end{bmatrix}$$

$$\Rightarrow P(x).P(y) = \begin{bmatrix} \cos x \cdot \cos y - \sin x \cdot \sin y & \cos x \sin y + \sin x \cdot \cos y \\ -\sin x \cos y - \cos x \cdot \sin y & -\sin x \cdot \sin y + \cos x \cdot \cos y \end{bmatrix}$$

$$\Rightarrow P(x).P(y) = \begin{bmatrix} \cos(x+y) & \sin(x+y) \\ -\sin(x+y) & \cos(x+y) \end{bmatrix}$$

now,

$$P(y).P(x) = \begin{bmatrix} \cos y & \sin y \\ -\sin y & \cos y \end{bmatrix} \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$$

$$\Rightarrow P(y).P(x) = \begin{bmatrix} \cos y \cdot \cos x - \sin y \cdot \sin x & \cos y \sin x + \sin y \cdot \cos x \\ -\sin y \cos x - \cos y \cdot \sin x & -\sin y \cdot \sin x + \cos y \cdot \cos x \end{bmatrix}$$

$$\Rightarrow P(y).P(x) = \begin{bmatrix} \cos(x+y) & \sin(x+y) \\ -\sin(x+y) & \cos(x+y) \end{bmatrix}$$

$$\therefore P(x).P(y) = P(x+y) = P(y).P(x)$$

47. If  $A$  is square matrix such that  $A^2 = A$ , then show that  $(I + A)^3 = 7A + I$ .

**Ans: Given:** Matrix  $A$  and  $A^2 = A$ .

Use  $(I + A)^2 = (I + A)(I + A)$ . Then, use matrix algebra.

Here,  $A^2 = A$

$$(I + A)(I + A) = I^2 + IA + AI + A^2$$

$$\Rightarrow (I + A)(I + A) = I^2 + 2AI + A^2$$

$$(I + A)(I + A) = I^2 + 2AI + A$$

$$(I + A)(I + A) = I + 3A$$

Now,

$$(I + A)(I + A)(I + A) = (I + A)(I + 3A)$$

$$\Rightarrow (I + A)(I + A)(I + A) = I^2 + 3AI + AI + 3A^2$$

$$\Rightarrow (I + A)(I + A)(I + A) = I^2 + 4AI + 3A$$

$$\Rightarrow (I + A)(I + A)(I + A) = I + 7A$$

$$\Rightarrow (I + A)^3 = 7A + I$$

**48. If  $A, B$  are square matrices of same order and  $B$  is a skew symmetric matrix, then show that  $A'BA$  is skew-symmetric.**

**Ans: Given:** Matrices  $A, B$  and  $B' = -B$ .

Use transpose identities of matrix.

$$(A'BA)' = A'B'A$$

$$\Rightarrow (A'BA)' = A'(-B)A$$

$$\Rightarrow (A'BA)' = -A'BA$$

$\therefore A'BA$  is skew-symmetric.

**Long Answer Type Questions:**

**49. If  $AB = BA$  for any two square matrices, then prove by mathematical induction that  $(AB)^n = A^nB^n$ .**

**Ans: Given:**  $AB = BA$ .

Consider  $P(n)$  and  $P(k)$  be true. Then, prove  $P(k + 1)$  is true.

$$P(n) : (AB)^n = A^n B^n$$

$$P(1) : (AB)^1 = A^1 B^1$$

$$\Rightarrow P(1) : AB = AB$$

$\therefore P(1)$  is true.

Now,

$$P(k) : (AB)^k = A^k B^k$$

$\therefore P(k)$  is true.

also,

$$P(k + 1) : (AB)^{k+1} = A^{k+1} B^{k+1}$$

$$P(k + 1) : (AB)^k (AB) = A^{k+1} B^{k+1}$$

$$P(k + 1) : A^k B^k (BA) = A^{k+1} B^{k+1}$$

$$P(k + 1) : A^{k+1} . B^{k+1} = A^{k+1} B^{k+1}$$

$$P(k + 1) : (AB)^{k+1} = A^{k+1} B^{k+1}$$

$\therefore P(k + 1)$  is true for all  $n \in \mathbb{N}$ , where  $P(k)$  is true.

By mathematical induction  $(AB)^n = A^n B^n$  is true for all  $n \in \mathbb{N}$ .

50. Find  $x, y$  and  $z$ , if  $A = \begin{bmatrix} 0 & 2y & z \\ x & y & -z \\ x & -y & z \end{bmatrix}$  satisfies  $A' = A^{-1}$ .

**Ans: Given:** Matrix  $A$  and  $A' = A^{-1}$ .

First, find transpose matrix. Then, use  $AA' = I$ .

$$A = \begin{bmatrix} 0 & 2y & z \\ x & y & -z \\ x & -y & z \end{bmatrix} \text{ and } A' = \begin{bmatrix} 0 & x & x \\ 2y & y & -y \\ z & -z & z \end{bmatrix}$$

$$A' = A^{-1}$$

$$\Rightarrow AA' = AA^{-1}$$

$$\Rightarrow AA' = I$$

$$\Rightarrow \begin{bmatrix} 0 & 2y & z \\ x & y & -z \\ x & -y & z \end{bmatrix} \begin{bmatrix} 0 & x & x \\ 2y & y & -y \\ z & -z & z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4y^2 + z^2 & 2y^2 - z^2 & -2y^2 + z^2 \\ 2y^2 - z^2 & x^2 + y^2 + z^2 & x^2 - y^2 - z^2 \\ -2y^2 + z^2 & x^2 - y^2 - z^2 & x^2 + y^2 + z^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow 2y^2 - z^2 = 0 \text{ or } 2y^2 = z^2$$

$$\Rightarrow 4y^2 + z^2 = 1$$

$$\Rightarrow 2z^2 + z^2 = 1$$

$$\Rightarrow z = \pm \frac{1}{\sqrt{3}}$$

$$\therefore y^2 = \frac{z^2}{2}$$

$$\Rightarrow y = \pm \frac{1}{\sqrt{6}}$$

now,

$$x^2 + y^2 + z^2 = 1$$

$$\Rightarrow x^2 = 1 - y^2 - z^2$$

$$\Rightarrow x^2 = 1 - \frac{1}{6} - \frac{1}{3}$$

$$\Rightarrow x = \pm \frac{1}{\sqrt{2}}$$

$$\therefore x = \pm \frac{1}{\sqrt{2}}, y = \pm \frac{1}{\sqrt{6}} \text{ and } z = \pm \frac{1}{\sqrt{3}}.$$

**51. If possible, using elementary row transformations, find the inverse of the following matrices.**

$$(i) \begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix}$$

**Ans: Given:** Matrices.

For inverse of matrices use  $A = IA$ .

Here,

$$A = IA$$

$$\Rightarrow \begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\Rightarrow \begin{bmatrix} 2 & -1 & 3 \\ -3 & 2 & 4 \\ -3 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \quad [R_2 \rightarrow R_2 + R_1]$$

$$\Rightarrow \begin{bmatrix} 2 & -1 & 3 \\ -3 & 2 & 4 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} A \quad [R_3 \rightarrow R_3 - R_2]$$

$$\Rightarrow \begin{bmatrix} -1 & 1 & 7 \\ -3 & 2 & 4 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} A \quad [R_2 \rightarrow R_2 + R_1]$$

$$\Rightarrow \begin{bmatrix} -1 & 1 & 7 \\ 0 & -1 & -17 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ -5 & -2 & 0 \\ -1 & -1 & 1 \end{bmatrix} A \quad [R_2 \rightarrow R_2 - 3R_1]$$

$$\Rightarrow \begin{bmatrix} -1 & 0 & -10 \\ 0 & -1 & -17 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -1 & 0 \\ -5 & -2 & 0 \\ 1 & 1 & -1 \end{bmatrix} A \quad [R_1 \rightarrow R_1 + R_2 \text{ and } R_3 \rightarrow -1.R_3]$$

$$\Rightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 9 & -10 \\ 12 & 15 & -17 \\ 1 & 1 & -1 \end{bmatrix} A \quad [R_1 \rightarrow R_1 + 10R_3 \text{ and } R_2 \rightarrow R_2 + 17R_3]$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -7 & -9 & 10 \\ -12 & -15 & 17 \\ 1 & 1 & -1 \end{bmatrix} A \quad [R_1 \rightarrow -1.R_1 \text{ and } R_2 \rightarrow -1.R_2]$$

$$\therefore \text{Inverse of } A = \begin{bmatrix} -7 & -9 & 10 \\ -12 & -15 & 17 \\ 1 & 1 & -1 \end{bmatrix}.$$

$$(ii) \begin{bmatrix} 2 & 3 & -3 \\ -1 & -2 & 2 \\ 1 & 1 & -1 \end{bmatrix}$$

**Ans: Given:** Matrices.

For inverse of matrices use  $A = IA$ .

Here,

$$A = IA$$

$$\Rightarrow \begin{bmatrix} 2 & 3 & -3 \\ -1 & -2 & 2 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\Rightarrow \begin{bmatrix} 0 & 1 & -1 \\ 0 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} A \quad [R_2 \rightarrow R_2 + R_3 \text{ and } R_1 \rightarrow R_1 - 2R_3]$$

$$\Rightarrow \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} A \quad [R_2 \rightarrow R_2 + R_1]$$

$$(iii) \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

**Ans: Given:** Matrices.

For inverse of matrices use  $A = IA$ .

Here,

$$A = IA$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 3 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \quad [R_2 \rightarrow R_2 - R_1]$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} A \quad [R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 + R_1]$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 0 & 1 & \frac{5}{2} \\ 4 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{5}{2} & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} A \quad [R_3 \rightarrow R_3 + R_1 \text{ and } R_2 \rightarrow R_2 - \frac{1}{2}R_1]$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{5}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \quad [R_3 \rightarrow R_3 - 2R_1]$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{5}{2} & 1 & 0 \\ \frac{5}{2} & -1 & 1 \end{bmatrix} A \quad [R_3 \rightarrow R_3 - R_2]$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & \frac{-1}{2} \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -\frac{5}{2} & 1 & 0 \\ 5 & -2 & 2 \end{bmatrix} A \quad [R_1 \rightarrow \frac{1}{2}R_1 \text{ and } R_3 \rightarrow 2R_3]$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & 5 \\ 5 & -2 & -2 \end{bmatrix} A \quad [R_1 \rightarrow R_1 + \frac{1}{2}R_3 \text{ and } R_2 \rightarrow R_2 - \frac{5}{2}R_3]$$

$$\therefore \text{Inverse of } A \text{ is } \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & -2 \end{bmatrix}.$$

52. Express the matrix  $\begin{bmatrix} 2 & 3 & 1 \\ 1 & -1 & 5 \\ 4 & 1 & -2 \end{bmatrix}$  as the sum of symmetric and skew-symmetric matrix.

**Ans: Given:** Matrix A.

Sum of symmetric and skew-symmetric matrix is given by  $A = \frac{A + A'}{2} + \frac{A - A'}{2}$ .

Where  $\frac{A + A'}{2}$  is symmetric and  $\frac{A - A'}{2}$  is skew-symmetric matrix.

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & -1 & 5 \\ 4 & 1 & -2 \end{bmatrix} \text{ and } A' = \begin{bmatrix} 2 & 1 & 4 \\ 3 & -1 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$\Rightarrow \frac{A + A'}{2} = \frac{1}{2} \begin{bmatrix} 4 & 4 & 5 \\ 4 & -2 & 3 \\ 5 & 3 & 4 \end{bmatrix}$$

$$\Rightarrow \frac{A + A'}{2} = \begin{bmatrix} 2 & 2 & \frac{5}{2} \\ 2 & -1 & \frac{3}{2} \\ \frac{5}{2} & \frac{3}{2} & 2 \end{bmatrix}$$

also,

$$\frac{A - A'}{2} = \frac{1}{2} \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}$$

$$\Rightarrow \frac{A - A'}{2} = \begin{bmatrix} 0 & 1 & \frac{-3}{2} \\ -1 & 0 & \frac{1}{2} \\ \frac{3}{2} & \frac{-1}{2} & 0 \end{bmatrix}$$

$$\therefore \frac{A + A'}{2} + \frac{A - A'}{2} = \begin{bmatrix} 2 & 2 & \frac{5}{2} \\ 2 & -1 & \frac{3}{2} \\ \frac{5}{2} & \frac{3}{2} & 2 \end{bmatrix} + \begin{bmatrix} 0 & 1 & \frac{-3}{2} \\ -1 & 0 & \frac{1}{2} \\ \frac{3}{2} & \frac{-1}{2} & 0 \end{bmatrix}.$$

### Objective Type Questions:

53. The matrix  $P = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 0 \end{bmatrix}$  is a

- (A) Square matrix
- (B) Diagonal matrix
- (C) Unit matrix
- (D) None of these

**Ans: Given:** Matrix P.

Matrix can be square, diagonal, unit depending on the order and elements of matrix.

Since, the order of matrix P is  $3 \times 3$ . The number of rows and columns are same.

Therefore, matrix P is square matrix.

**Correct Answer: A**

54. Total number of possible matrices of order  $3 \times 3$  with each entry 2 or 0 is,

- (A) 9
- (B) 27
- (C) 81
- (D) 512

**Ans: Given:** Order of matrix.

Total number of elements in matrices is  $n$ . Then, total number of matrices with any two entries will be  $2^n$ .

Therefore, total number of possible matrices of order  $3 \times 3$  with each entry 2 or 0 is  $2^9$  or 512.

**Correct Answer: D**

55.  $\begin{bmatrix} 2x + y & 4x \\ 5x - 7 & 4x \end{bmatrix} = \begin{bmatrix} 7 & 7y - 13 \\ y & x + 6 \end{bmatrix}$ , then the value of  $x + y$  is

- (A)  $x = 3, y = 1$
- (B)  $x = 2, y = 3$
- (C)  $x = 2, y = 4$
- (D)  $x = 3, y = 3$

**Given:** Matrix equation.

Use equality of two matrices. Then, compare the elements.

Here,

$$4x = x + 6 \text{ or } x = 2$$

also,

$$4x = 7y - 13$$

$$\Rightarrow y = 3$$

$$\therefore x + y = 2 + 3$$

$$x + y = 5$$

56. If  $A = \frac{1}{\pi} \begin{bmatrix} \sin^{-1}(x\pi) & \tan^{-1}\left(\frac{x}{\pi}\right) \\ \sin^{-1}\left(\frac{x}{\pi}\right) & \cot^{-1}(\pi x) \end{bmatrix}$  and  $B = \frac{1}{\pi} \begin{bmatrix} -\cos^{-1}(x\pi) & \tan^{-1}\left(\frac{x}{\pi}\right) \\ \sin^{-1}\left(\frac{x}{\pi}\right) & -\tan^{-1}(\pi x) \end{bmatrix}$ ,

then  $A - B$  is equal to

(A)  $I$

(B)  $0$

(C)  $2I$

(D)  $\frac{1}{2}I$

**Ans:** Use matrix algebra to find difference of two matrices.

**Correct Answer:** D

57. If  $A$  and  $B$  are two matrices of the order

$3 \times m$  and  $3 \times n$ , respectively and  $m = n$ , then order of matrix  $(5A - 2B)$  is

(A)  $m \times 3$

(B)  $3 \times 3$

(C)  $m \times n$

(D)  $3 \times n$

**Ans: Given:** Order of matrices A and B.

Matrix of same order can be added or subtracted.

For,  $m = n$ . The order of matrix A and B can be  $3 \times m$  or  $3 \times n$ .

Therefore, the order of matrix  $(5A - 2B)$  is  $3 \times n$ .

**Correct Answer:** D

58. If  $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ , then  $A^2$  is equal to

(A)  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(B)  $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

(C)  $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$

(D)  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

**Ans: Given:** Matrix A.

Use matrix multiplication  $A^2 = A.A$ .

$$A^2 = A.A$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

**Correct Answer: D**

59. If matrix  $A = [a_{ij}]_{2 \times 2}$ , where  $a_{ij} = 1$ , if  $i \neq j = 0$  and if  $i = j$ , then  $A^2$  is equal to

(A) I

(B) A

(C) 0

(D) None of these

**Ans: Given:** Matrix  $A = [a_{ij}]_{2 \times 2}$ .

Consider a matrix A according to given conditions. Then, use  $A^2 = A.A$ .

Let,

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A^2 = A.A$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = I$$

**Correct Answer: A**

60. The matrix  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$  is a

- (A) Identity matrix
- (B) Symmetric matrix
- (C) Skew-symmetric matrix
- (D) None of these

**Ans: Given:** Matrix.

Take transpose of matrix. For symmetric matrix  $A' = A$ .

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$A' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\Rightarrow A' = A.$$

**Correct Answer: B**

61. The matrix  $\begin{bmatrix} 0 & -5 & 8 \\ 5 & 0 & 12 \\ -8 & -12 & 0 \end{bmatrix}$  is a

- (A) Diagonal matrix

**(B) Symmetric matrix**

**(C) Skew-symmetric matrix**

**(D) Scalar matrix**

**Ans: Given:** Matrix.

Take transpose of matrix. For skew-symmetric matrix  $A' = -A$ .

$$B = \begin{bmatrix} 0 & -5 & 8 \\ 5 & 0 & 12 \\ -8 & -12 & 0 \end{bmatrix}$$

$$B' = \begin{bmatrix} 0 & 5 & -8 \\ -5 & 0 & -12 \\ 8 & 12 & 0 \end{bmatrix}$$

$$\Rightarrow B' = -B.$$

**Correct Answer:** C

**62. If A is matrix of order  $m \times n$  and B is a matrix such that  $AB'$  and  $B'A$  are both defined, then order of matrix B is**

**(A)  $m \times m$**

**(B)  $n \times n$**

**(C)  $n \times m$**

**(D)  $m \times n$**

**Ans: Given:** Matrix A and B.

Consider matrix B with order  $p \times q$ . Then, check where  $AB'$  and  $B'A$  are defined.

$$A = [a_{ij}]_{m \times n} \text{ and } B = [b_{ij}]_{p \times q}$$

$$\therefore B' = [b_{ij}]_{q \times p}$$

for  $n = q$ ,  $AB'$  is defined.

for  $p = m$ ,  $B'A$  is defined.

$\therefore$  Order of matrix  $B$  is  $m \times n$ .

**Correct Answer: D**

**63. If  $A$  and  $B$  are matrices of same order, then  $(AB' - BA')$  is a**

**(A) Skew-symmetric matrix**

**(B) Null matrix**

**(C) Symmetric matrix**

**(D) Unit matrix**

**Ans: Given:** Matrices  $A$  and  $B$  of same order.

Take transpose of the matrix  $(AB' - BA')$ .

$$X = (AB' - BA')$$

$$\therefore X' = (AB' - BA)'$$

$$\Rightarrow X' = (AB')' - (BA)'$$

$$\Rightarrow X' = (B')'A' - (A)'B'$$

$$\Rightarrow X' = BA' - AB'$$

$$\Rightarrow X' = -(AB' - BA')$$

$$\Rightarrow X' = -X$$

$\therefore (AB' - BA')$  is skew-symmetric matrix.

**Correct Answer: A**

64. If  $A$  is a square matrix such that  $A^2 = I$ , then  $(A - I)^3 + (A + I)^3 - 7A$  is equal to

- (A)  $A$
- (B)  $I - A$
- (C)  $I + A$
- (D)  $3A$

**Ans: Given:** Matrix  $A$  and  $A^2 = I$ .

Use  $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$ .

$$\begin{aligned}
 (A - I)^3 + (A + I)^3 - 7A &= [(A - I) + (A + I)\{(A - I)^2 + (A + I)^2 - (A - I)(A + I)\}] - 7A \\
 \Rightarrow (A - I)^3 + (A + I)^3 - 7A &= [(2A)\{A^2 + I^2 - 2AI + A^2 + I^2 + AI - (A^2 - I^2)\}] - 7A \\
 \Rightarrow (A - I)^3 + (A + I)^3 - 7A &= 2A[I + I^2 + I + I^2 - A^2 + I^2] - 7A \\
 \Rightarrow (A - I)^3 + (A + I)^3 - 7A &= 2A[5I - I] - 7A \\
 \Rightarrow (A - I)^3 + (A + I)^3 - 7A &= 8AI - 7AI \\
 \Rightarrow (A - I)^3 + (A + I)^3 - 7A &= AI \\
 \therefore (A - I)^3 + (A + I)^3 - 7A &= A
 \end{aligned}$$

**Correct Answer:** A

65. For any two matrices  $A$  and  $B$ , we have

- (A)  $AB = BA$
- (B)  $AB^{-1}BA$
- (C)  $AB = O$
- (D) None of these

**Ans: Given:** Matrices A and B.

First, find which conditions holds always true for matrices A and B.

The conditions  $AB = BA$ ,  $AB \neq BA$ ,  $AB = O$  are not always true for two matrices A and B.

**Correct Answer:** D

**66. On using elementary column operation  $C_2 \rightarrow C_2 - 2C_1$  in the following**

**matrix equation  $\begin{bmatrix} 1 & -3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$ , we have**

(A)  $\begin{bmatrix} 1 & -5 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$

(B)  $\begin{bmatrix} 1 & -5 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ 0 & 2 \end{bmatrix}$

(C)  $\begin{bmatrix} 1 & -5 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -2 & 4 \end{bmatrix}$

(D)  $\begin{bmatrix} 1 & -5 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ 2 & 0 \end{bmatrix}$

**Ans: Given:** Matrix equation.

Use the elementary column operation  $C_2 \rightarrow C_2 - 2C_1$  in matrix equation.

Applying  $C_2 \rightarrow C_2 - 2C_1$  on first product matrix of equation, we get

$$\begin{bmatrix} 1 & -5 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ 2 & 0 \end{bmatrix}$$

**Correct Answer: D**

**67. On using elementary row operation  $R_1 \rightarrow R_1 - 3R_2$  in the following matrix**

**equation  $\begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$ , we have**

(A)  $\begin{bmatrix} -5 & -7 \\ 3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -7 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

(B)  $\begin{bmatrix} -5 & -7 \\ 3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & -3 \\ 1 & 1 \end{bmatrix}$

(C)  $\begin{bmatrix} -5 & -7 \\ 3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & -7 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

(D)  $\begin{bmatrix} 4 & 2 \\ -5 & -7 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -3 & -3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

**Ans: Given:** Matrix equation.

Use the elementary row operation  $R_1 \rightarrow R_1 - 3R_2$  in matrix equation.

Applying  $R_1 \rightarrow R_1 - 3R_2$  on first product matrix of equation, we get

$$\begin{bmatrix} -5 & -7 \\ 3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -7 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

**Correct Answer: A**

**Fill in the blanks:**

**68. \_\_\_\_\_ matrix is both symmetric and skew-symmetric matrix.**

**Ans: Given:** Matrix is both symmetric and skew-symmetric matrix.

For symmetric matrix  $A' = A$  and for skew-symmetric matrix  $A' = -A$ .

Consider a null matrix,

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow A' = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ or } - \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore A' = A \text{ and } A' = -A.$$

Therefore, null matrix is both symmetric and skew-symmetric matrix.

**69. Sum of two skew-symmetric matrices is always ..... Matrix.**

**Ans:** Sum of two skew-symmetric matrices is always skew-symmetric matrix.

Consider two matrices A and B. Then,

$$A' = -A \text{ and } B' = -B$$

$$\Rightarrow -A - B = -(A + B).$$

**70. The negative of matrix is obtained by multiplying it by .....**

**Ans:** Consider a matrix A. The negative of matrix is obtained by multiplying it by  $-1$ .

$$\therefore -A = -1[A].$$

**71. The product of any matrix by the scalar ..... is the null matrix.**

**Ans:** Consider a matrix,  $A = \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}$ .

$$\Rightarrow 0.A = 0 \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix} = 0.$$

The product of any matrix by the scalar 0 is the null matrix.

**72. A matrix which is not a square matrix is called a ..... Matrix.**

**Ans:** Consider a matrix,  $A = \begin{bmatrix} 1 & 0 & 2 \end{bmatrix}$ .

The matrix which is not square is considered as rectangular matrix.

**73. Matrix multiplication is ..... over addition**

**Ans:** Consider three matrices A, B and C.

$$1. A(B + C) = AB + AC$$

$$2. (A + B)C = AC + BC$$

Therefore, matrix multiplication is distributive over addition.

**74. If A is symmetric matrix, then  $A^3$  is a ..... matrix.**

**Ans: Given:** Symmetric matrix A.

For symmetric matrix  $A' = A$  and for skew-symmetric matrix  $A' = -A$ .

Here,

$$A' = A$$

now,  $A^3$

$$(A^3)' = (A')^3$$

$$\Rightarrow (A^3)' = A^3$$

Therefore,  $A^3$  is a symmetric matrix.

**75. If  $A$  is a skew-symmetric matrix, then  $A^2$  is a .....**

**Ans: Given:**  $A' = -A$ .

For symmetric matrix  $A' = A$  and for skew-symmetric matrix  $A' = -A$ .

Here,

$$A' = -A$$

now,  $A^2$

$$(A^2)' = (A')^2$$

$$\Rightarrow (A^2)' = (-A)^2$$

$$\Rightarrow (A^2)' = A^2$$

Therefore,  $A^2$  is a symmetric matrix.

**76. If  $A$  and  $B$  are square matrices of same order, then**

**(i)  $(AB)' = \dots\dots\dots$**

**Ans: Given:** Matrices  $A$  and  $B$ .

Transpose the matrices. Then, use matrix identities.

Here,

$$(AB)' = B'A'$$

**(ii)  $(kA)' = \dots\dots(k \text{ is any scalar})$**

**Ans: Given:** Matrices A and B.

Transpose the matrices. Then, use matrix identities.

Here,

$$(kA)' = kA'$$

**(iii)  $[k(A - B)]' = \dots\dots$**

**Ans: Given:** Matrices A and B.

Transpose the matrices. Then, use matrix identities.

Here,

$$(AB)' = B'A'$$

Here,

$$[k(A - B)]' = k(A - B)'$$

$$\Rightarrow [k(A - B)]' = k(A' - B')$$

**77. If A is a skew-symmetric matrix, then  $kA$  is a  $\dots\dots$  (where, k is any scalar).**

**Ans: Given:**  $A' = -A$ .

For skew-symmetric matrix  $A' = -A$  and for symmetric matrix  $A' = A$ .

Here,

$$A' = -A$$

also,  $kA$

$$(kA)' = kA'$$

$$\Rightarrow (kA)' = kA'$$

$$\Rightarrow (kA)' = k(-A)$$

$$\Rightarrow (kA)' = -(kA)$$

Therefore,  $kA$  is skew-symmetric matrix.

**78. If  $A$  and  $B$  are symmetric matrices, then**

**(i)  $AB - BA$  is a .....**

**Ans: Given:** Matrices  $A$  and  $B$ ,  $A' = A$  and  $B' = B$ .

For skew-symmetric matrix  $A' = -A$  and for symmetric matrix  $A' = A$ .

Here,

$$(AB - BA)' = (AB)' - (BA)'$$

$$\Rightarrow (AB - BA)' = B'A' - A'B'$$

$$\Rightarrow (AB - BA)' = BA - AB$$

$$\Rightarrow (AB - BA)' = -(AB - BA)$$

Therefore,  $AB - BA$  is a skew-symmetric matrix.

**(ii)  $BA - 2AB$  is a .....**

**Ans: Given:** Matrices  $A$  and  $B$ ,  $A' = A$  and  $B' = B$ .

For skew-symmetric matrix  $A' = -A$  and for symmetric matrix  $A' = A$ .

Here,

$$\begin{aligned}(BA - 2AB)' &= (BA)' - 2(AB)' \\ \Rightarrow (BA - 2AB)' &= A'B' - 2B'A' \\ \Rightarrow (BA - 2AB)' &= AB - 2BA \\ \Rightarrow (BA - 2AB)' &= -(2BA - AB)\end{aligned}$$

Therefore,  $BA - 2AB$  is neither skew-symmetric nor symmetric matrix.

**79. If  $A$  is symmetric matrix, then  $B'AB$  is .....**

**Ans: Given:**  $A' = A$ .

For skew-symmetric matrix  $A' = -A$  and for skew symmetric matrix  $A' = -A$ .

Here,

$$\begin{aligned}(B'AB)' &= [B'(AB)]' \\ \Rightarrow (B'AB)' &= (AB)'(B')' \\ \Rightarrow (B'AB)' &= (B'A')B \\ \Rightarrow (B'AB)' &= B'AB\end{aligned}$$

Therefore,  $B'AB$  is a symmetric matrix.

**80. If  $A$  and  $B$  are symmetric matrices of same order, then  $AB$  is symmetric if and only if .....**

**Ans: Given:**  $A$  and  $B$  are two skew symmetric matrices.

For symmetric matrix  $A' = A$  and for skew symmetric matrix  $A' = -A$ .

Take transpose,

$$(AB)' = B'A'$$

$$\Rightarrow (AB)' = (-B)(-A)$$

$$(AB)' = BA$$

if,  $AB = BA$ , then

$$(AB)' = AB.$$

For  $AB = BA$ ,  $AB$  is symmetric matrix.

**81. In applying one or more row operations while finding  $A^{-1}$  by elementary row operations, we obtain all zeroes in one or more, then  $A^{-1}$  .....**

**Ans:** While solving the inverse of matrix, if we obtain all zeroes in one or more row. Then, inverse of the matrix does not exist.

Therefore,  $A^{-1}$  does not exist.

**True/false:**

**82. A matrix denotes a number.**

**Ans:** False.

A matrix is defined as the ordered rectangular array which contains numbers or functions.

**83. Matrices of any order can be added.**

**Ans:** False.

According to matrix algebra, two matrices can be added only when they have same order.

**84. Two matrices are equal, if they have same number of rows and same number of columns.**

**Ans:** False.

The two matrices are said to be equal only when they have same elements and same order. Matrices with same number of rows and same number of columns are square matrices.

**85. Matrices of different order cannot be subtracted.**

**Ans:** True.

According to matrix algebra, two matrices of same order can be added or subtracted. Therefore, matrices of different order cannot be subtracted.

**86. Matrix addition is associative as well as commutative.**

**Ans:** True.

According to properties of matrices, Matrix addition is both associative and commutative.

$$(A + B) + C = A + (B + C) \text{ and } A + B = B + A.$$

**87. Matrix multiplication is commutative.**

**Ans:** False.

According to matrix multiplication, if  $AB$  and  $BA$  are both defined then,  $AB \neq BA$ . Therefore, matrix multiplication is not commutative.

**88. A square matrix where every element is unity is called an identity matrix.**

**Ans:** False.

A matrix is said to be an identity matrix if and only if every diagonal element of matrix is unity. Therefore, square matrix where every element is unity is not an identity matrix.

**89. If  $A$  and  $B$  are two square matrices of the same order, then  $A + B = B + A$ .**

**Ans:** According to properties of matrices, matrix addition of two matrices with same order is commutative.

$A + B = B + A$ ,  
where  $A$  and  $B$  are square matrices.

**90. If  $A$  and  $B$  are two matrices of same order, then  $A - B = B - A$ .**

**Ans:** False.

According to properties of matrices, addition of two matrices is commutative.

$$\begin{aligned}\therefore A + (-B) &= A - B \\ \Rightarrow A - B &= -(B - A) \\ \Rightarrow -(B - A) &\neq B - A\end{aligned}$$

**91. If matrix  $AB = 0$ , then  $A = 0$  or  $B = 0$  or both  $A$  and  $B$  are null matrices.**

**Ans:** False.

For two non-zero matrices A and B of same order. It is possible that  $A.B = 0$ .  
Therefore,  $A.B = 0$  does not implies that  $A = 0$  or  $B = 0$ .

**92. Transpose of column matrix is a column matrix.**

**Ans:** False.

Consider a column matrix  $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$  and  $A' = [1 \ 2 \ 3]$ .

Therefore, transpose of column matrix is a row matrix.

**93. If A and B are two square matrices of the same order, then  $AB = BA$ .**

**Ans:** False.

Since, matrix multiplication does not follow commutative law. It is not always true that  $AB = BA$ .

**94. If each of the three matrices of the same order are symmetric, then their sum is a symmetric matrix.**

**Ans:** True.

Consider three matrices A, B and C of same order.

$$\therefore A' = A, B' = B \text{ and } C' = C.$$

$$(A + B + C)' = A' + B' + C'$$

$$\Rightarrow (A + B + C)' = A + B + C$$

Therefore,  $(A + B + C)$  is symmetric matrix.

**95.** If  $A$  and  $B$  are any two matrices of the same order, then  $(AB)' = A'B'$ .

**Ans:** False.

According to transpose properties of matrices, the transpose of matrix  $AB$  is  $(AB)' = B'A'$ .

**96.** If  $(AB)' = B'A'$ , where  $A$  and  $B$  are not square matrices, then number of rows in  $A$  is equal to number of columns in  $B$  and number of columns in  $A$  is equal to number of columns in  $B$ .

**Ans: Given:**  $(AB)' = B'A'$ .

For multiplication of non-square matrices the product must be defined as per the order of matrix.

True.

Let, matrices  $A$  and  $B$  with order  $m \times n$  and  $p \times q$  respectively.

As,  $(AB)' = B'A'$

$\therefore A_{m \times n} B_{p \times q}$  is defined for  $n = p$ .

order of  $(AB)$  is  $m \times q$ .

$\Rightarrow (AB)'$  is of order  $q \times m$ .

$\Rightarrow B'A'$  is of order of  $q \times m$ .

$\therefore (AB)' = B'A'$  is true.

**97.** If  $A, B$  and  $C$  are square matrices of same order, then  $AB = AC$  always implies that  $B = C$ .

**Ans: Given:**  $A, B$  and  $C$  are square matrices.

Consider three matrices such that  $AB = AC$ . Then, check for equality for  $B$  and  $C$ .

False.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 1 & 3 \end{bmatrix} \text{ and } C = \begin{bmatrix} 0 & 0 \\ 3 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$AC = \begin{bmatrix} 0 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$\therefore AB = AC$  but  $B \neq C$ .

**98.  $AA'$  is always a symmetric matrix for any matrix  $A$ .**

**Ans: Given:** Matrix  $A$ .

For skew-symmetric matrix  $A' = -A$  and for symmetric matrix  $A' = A$ .

True.

For  $AA'$ ,

$$(AA')' = (A')'A'$$

$$\Rightarrow (AA')' = (AA')$$

Therefore,  $AA'$  is always symmetric matrix.

**99. If  $A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 4 & 2 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$ , then  $AB$  and  $BA$  are defined and equal.**

**Ans: Given:** Matrices  $A$  and  $B$ .

Use matrix multiplication  $A.B$  and  $B.A$ . Then, check equality of  $AB$  and  $BA$ .

False.

$$AB = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 4 & 2 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}_{3 \times 2}$$

AB is defined,

$$\Rightarrow AB = \begin{bmatrix} 14 & 20 \\ 22 & 25 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 2 & 3 & -1 \\ 1 & 4 & 2 \end{bmatrix}_{2 \times 3}$$

BA is defined,

$$\Rightarrow BA = \begin{bmatrix} 7 & 18 & 4 \\ 13 & 32 & 6 \\ 5 & 10 & 0 \end{bmatrix}$$

$\therefore AB \neq BA$ .

**100. If A is skew symmetric matrix, then  $A^2$  is a symmetric matrix.**

**Ans: Given:**  $A' = -A$ .

For symmetric matrix  $A' = A$  and for skew-symmetric matrix  $A' = -A$ .

True.

Here,

$$A' = -A$$

now,  $A^2$

$$(A^2)' = (A')^2$$

$$\Rightarrow (A^2)' = (-A)^2$$

$$\Rightarrow (A^2)' = A^2$$

Therefore,  $A^2$  is a symmetric matrix.

**101.  $(AB)^{-1} = A^{-1}B^{-1}$ , where A and B are invertible matrices satisfying commutative property with respect to multiplication.**

**Ans: Given:**  $(AB)^{-1} = A^{-1}B^{-1}$ .

Use commutative property  $AB = BA$  of matrix multiplication.

True.

A and B are invertible matrices,

$$(AB)^{-1} = (BA)^{-1}$$

$$(AB)^{-1} = (AB)^{-1} \quad \{\text{using, } AB = BA\}$$

$$\Rightarrow B^{-1}A^{-1} = A^{-1}B^{-1}$$